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A comprehensive survey of large language models and multimodal large language models in medicine

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ABSTRACT

Since the release of ChatGPT and GPT-4, large language models (LLMs) and multimodal large language models (MLLMs) have attracted widespread attention for their exceptional capabilities in understanding, reasoning, and generation, introducing transformative paradigms for integrating artificial intelligence into medicine. This survey provides a comprehensive overview of the development, principles, application scenarios, challenges, and future directions of LLMs and MLLMs in medicine. Specifically, it begins by examining the paradigm shift, tracing the transition from traditional models to LLMs and MLLMs, and highlighting the unique advantages of these LLMs and MLLMs in medical applications. Next, the survey reviews existing medical LLMs and MLLMs, providing detailed guidance on their construction and evaluation in a clear and systematic manner. Subsequently, to underscore the substantial value of LLMs and MLLMs in healthcare, the survey explores five promising applications in the field. Finally, the survey addresses the challenges confronting medical LLMs and MLLMs and proposes practical strategies and future directions for their integration into medicine. In summary, this survey offers a comprehensive analysis of the technical methodologies and practical clinical applications of medical LLMs and MLLMs, with the goal of bridging the gap between these advanced technologies and clinical practice, thereby fostering the evolution of the next generation of intelligent healthcare systems.

1. Introduction

The introduction of the Transformer [1] has revolutionized the fields of Natural Language Processing (NLP) and Computer Vision (CV). The Transformer's robust parallel computing capabilities and self-attention mechanism facilitate the integration of vast training datasets, forming the foundation for LLM and MLLM development. A variety of Transformer-based LLMs and MLLMs have emerged to date, with this survey primarily focusing on the vision-language modality. Notable examples include the PaLM series [2,3], GPT series [4,5], and LLaMA series [6–8] among LLMs, and Gemini [9], GPT-4 [10], and LLaVA [11] among MLLMs. Their exceptional capabilities in understanding, reasoning, and generation have enabled them to achieve state-of-the-art performance across various downstream tasks, including text generation, machine translation, and visual question answering (VQA). LLMs and MLLMs exhibit increasingly robust generalization abilities, significantly impacting the medical domain and accelerating the convergence of artificial intelligence and medicine [12]. Notably, Google's Med-PaLM 2 [13] scored 86.5 on the United States Medical Licensing Examination (USMLE) [14], achieving expert-level performance [15] and further highlighting the immense potential of LLMs

in medicine. Additionally, emerging medical LLMs and MLLMs, such as ChatDoctor [16], ChatCAD [17], and LLaVA-Med [18], represent novel opportunities enabled by artificial intelligence in the medical field. These models offer promising solutions for medical report generation [19,20], clinical diagnosis [17,21], mental health services [22,23], and various other clinical applications.

Despite the significant academic breakthroughs of LLMs and MLLMs in the medical field, hospitals still face notable challenges in training their own medical LLMs and MLLMs and deploying them in practical clinical applications. First, training medical LLMs and MLLMs requires a substantial amount of medical data, which is costly to obtain, demands annotation by medical experts, and raises significant concerns about data privacy [24]. Second, the parameters and computational demands of LLMs and MLLMs are substantial, necessitating significant computational resources for their training and deployment [25,26], substantially increasing the adoption barrier for hospitals. Third, given the unique characteristics of the medical field, it is essential not only to evaluate the performance of LLMs and MLLMs on benchmarks but also to assess their instruction-following ability [5,11,27], safety, and

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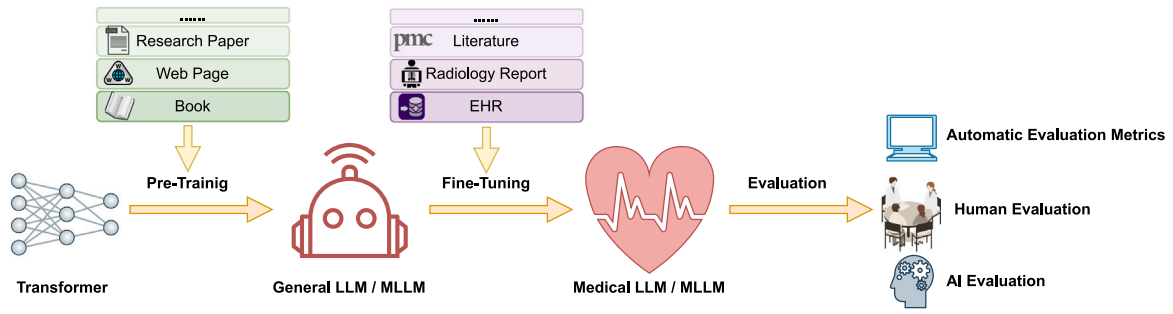


Fig. 1. The process of constructing and evaluating medical LLMs and MLLMs.

ethical considerations, necessitating additional training and evaluation strategies to enhance and measure the models' performance across multiple dimensions. Furthermore, the development of LLMs and MLLMs in the medical field is still in its infancy, with many of their potential application scenarios remaining undefined. Moreover, they face a range of challenges, including hallucinations [28–30] and a lack of up-to-date information [12], which significantly impede their practical use in clinical settings.

To address the aforementioned challenges, this survey begins by tracing the evolution of LLMs and MLLMs through the lens of paradigm shifts. Subsequently, it reviews existing medical LLMs and MLLMs, summarizing their structural characteristics. The survey then gathers datasets suitable for training medical LLMs and MLLMs and elaborates on methods for training and evaluating these models, as shown in Fig. 1. Furthermore, to highlight the significant potential impact of LLMs and MLLMs in medicine, this survey summarizes their applications in clinical practice and analyzes current limitations and potential solutions. Finally, the survey explores the future directions of medical LLMs and MLLMs, offering forward-looking and insightful perspectives.

Medicine is a multimodal field [31,32], making the study of medical MLLMs particularly important, as they can integrate and analyze information from various modalities to enhance clinical decision support, disease diagnosis, and treatment planning. However, the articles relevant to this survey mainly focus on medical LLMs and lack a detailed examination of medical MLLMs [15,33,34]. Additionally, most articles focus on the applications and impacts of LLMs in medicine but lack detailed discussions of technical aspects [12,26,35–37], such as datasets, model architectures, and construction methods. In contrast, this survey not only examines the background and principles of LLMs and MLLMs but also explores their applications and impacts in medicine, offering a clear logical structure and substantial depth and breadth. In summary, the contributions of this survey are as follows:

- This survey offers a thorough overview of medical LLMs and MLLMs, starting with an examination of their developmental background and architectural frameworks. Building on this foundation, it catalogs existing medical LLMs and MLLMs while providing a detailed analysis of their structural variations and key components.
- This survey systematically elucidates the complete process of medical LLMs and MLLMs, from training to evaluation, covering fine-tuning methods, evaluation strategies, and relevant medical datasets. Additionally, it highlights how to select appropriate datasets, fine-tuning methods, and evaluation strategies to assist researchers in the rapid development of medical LLMs and MLLMs.
- This survey summarizes the applications, challenges, and potential solutions of medical LLMs and MLLMs in clinical practice, while providing a forward-looking analysis of future developmental trajectories. It seeks to offer visionary perspectives that inspire advancements in the field, benefiting medical professionals and researchers alike.

This survey aims to advance the development of LLMs and MLLMs for clinical medicine applications, thereby promoting deeper integration between artificial intelligence and healthcare. The structure of this survey is outlined in Fig. 2: Section 2 reviews the development background of LLMs and MLLMs. Section 3 describes the architectures of current LLMs and MLLMs and highlights the structural differences among them. Section 4 covers the datasets for training medical LLMs and MLLMs, along with an overview of training and evaluation methods. Section 5 examines current potential applications of medical LLMs and MLLMs. Section 6 discusses the challenges and limitations of LLMs and MLLMs in clinical settings and proposes potential solutions. Section 7 offers a forward-looking perspective on the future of medical LLMs and MLLMs. Finally, Section 8 concludes the survey. In summary, readers interested in the foundational knowledge and principles of medical LLMs and MLLMs should refer to Section 2 to Section 4, while those focused on applications, challenges, and future directions should consult Section 5 and Section 7.

2. Background of LLMs and MLLMs

This section focuses on paradigm shifts, categorizing the development of NLP into four distinct stages, as illustrated in Fig. 3: (1) Supervised Learning; (2) Unsupervised Pre-training and Fine-tuning; (3) Unsupervised Pre-training and Prompt; (4) Text-only to Multimodal. Recent research [38] underscores the importance of high-quality datasets for LLMs and MLLMs. Accordingly, we introduce (5) High-quality Data, which examines the shift from reliance on large-scale data to an emphasis on high-quality data.

2.1. Supervised learning

Supervised learning is a fundamental paradigm in machine learning that focuses on minimizing a loss function. The objective can be expressed as follows:

$$\arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n \mathcal{L}(f(x_i; \theta), y_i) + \lambda \Omega(\theta) \quad (1)$$

where the first term represents the empirical risk, and the second term represents the regularization term. In supervised learning, a model is trained to map input variables x to output variables y by minimizing the discrepancy between $f(x; \theta)$ and y , where θ denotes model parameters, x may consist of manually extracted features or raw text, and y represents supervision signals such as category labels, text, or other forms.

Before the advent of pre-training methods, the supervised learning paradigm dominated the NLP field. Early NLP relied heavily on feature engineering [39], requiring researchers to extract and select features from datasets to perform tasks like text classification [40] and machine translation [41]. The advent of deep learning [42] enabled end-to-end model training, shifting research focus from feature engineering to model architecture design, with CNN and LSTM models emerging as prominent approaches. The supervised learning era in NLP marked a shift from feature selection to model architecture design, signifying a transition from feature engineering to structure engineering.

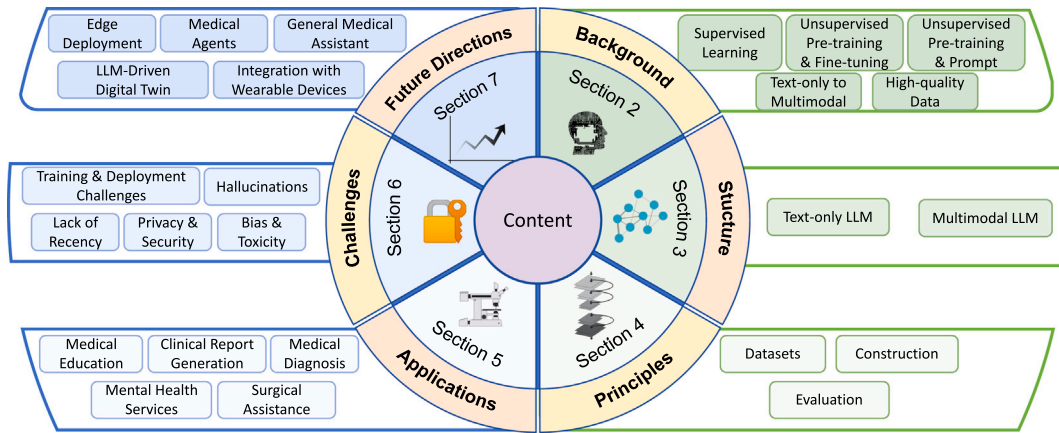


Fig. 2. The overall structure of the survey. Section 2 to Section 4 are biased toward principles of medical LLMs and MLLMs; Section 5 to Section 7 are biased toward practical clinical applications.

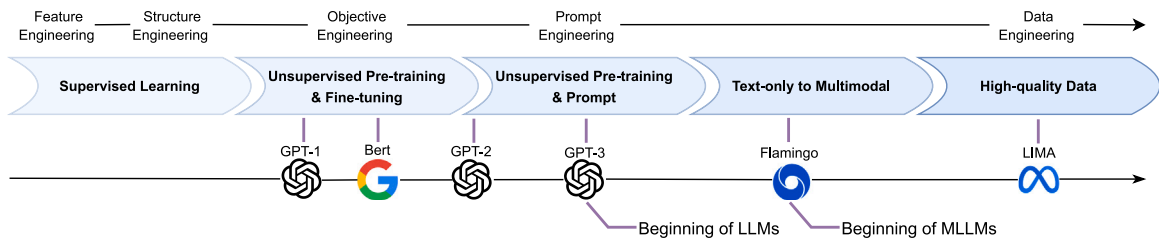


Fig. 3. Evolution of LLMs and MLLMs. The Evolution of LLMs and MLLMs. The upper section illustrates the research focuses and paradigm shifts across the evolution of these models, while the lower section highlights key milestones achieved at each stage.

2.2. Unsupervised pre-training and fine-tuning

Supervised learning depends on annotated datasets, which establish clear standards for model optimization. However, acquiring sufficient annotated data is challenging for certain tasks, particularly in medical domains, due to the scarcity of specialized annotators and the complexity of the annotation process [24]. The introduction of Transformer [1] revolutionized the NLP learning paradigm, rendering supervised learning increasingly marginalized [39].

Transformer-based models like GPT [43] and BERT [44] achieved state-of-the-art results through unsupervised pre-training on large-scale unlabeled text, followed by supervised fine-tuning (SFT) using task-specific objective functions. The emergence of GPT and BERT introduced a new NLP paradigm: unsupervised pre-training & fine-tuning. This paradigm also revolutionized language model development in the medical field, giving rise to prominent models like PubMedBERT [45] and BioBERT [46]. Compared to earlier models, these approaches offer several advantages: (1) Pre-training data can be drawn from any unannotated text corpus, such as biomedical literature, mitigating challenges related to limited annotated data in medical domains; (2) Training on large-scale unlabeled data enables the acquisition of general and abstract language representations, improving generalization; (3) Fine-tuning for downstream tasks requires only task-specific objective functions, eliminating extensive architectural modifications and facilitating a transition from structural to objective engineering.

2.3. Unsupervised pre-training and prompt

While GPT and BERT achieved state-of-the-art results in tasks like machine translation, sentiment analysis, and question-answering (QA), they still required task-specific fine-tuning for each downstream task. To develop a general language model capable of handling diverse tasks without task-specific fine-tuning, Radford et al. [47] curated a dataset

of over 8 million documents, totaling 40 GB of text data, encompassing examples from multiple domains and tasks, and used it to train GPT-2. GPT-2 set state-of-the-art benchmarks on 7 out of 8 language modeling tasks without requiring task-specific fine-tuning.

To further enhance the generalization capabilities of language models, Brown et al. [4] scaled the model to 175 billion parameters and significantly expanded the training dataset. This resulted in GPT-3, which demonstrated a qualitative leap in performance, showcasing remarkable few-shot capabilities without requiring fine-tuning. GPT-3 could handle unfamiliar tasks based solely on provided examples, often achieving performance on par with fine-tuned state-of-the-art models. As a result, GPT-3 is widely regarded as the beginning of LLMs [35]. The proposal of GPT-3 further revolutionized NLP by shifting the paradigm from unsupervised pre-training & fine-tuning to unsupervised pre-training & prompt [39]. Such models can handle most tasks effectively using user prompts and contextual examples. For example, Flan-PaLM [13] achieved state-of-the-art performance on MedQA, MedMCQA, PubMedQA, and MMLU clinical benchmarks by employing advanced prompting strategies. However, these models are highly sensitive to user-provided prompts, which directly influence the quality of their responses. This sensitivity has spurred researchers to explore prompting strategies in depth [48], initiating a shift from objective engineering to prompt engineering.

2.4. Text-only to multimodal

Inspired by GPT-3, researchers have intensified efforts in developing LLMs, resulting in prominent works such as GLM-130B [49], PaLM [2, 3], and LLaMA [6–8]. However, these LLMs remain text-focused, and despite progress in multimodal research, they often require fine-tuning for new tasks [50,51] or lack text generation capabilities [52,53], which restricts their application scope. Inspired by few-shot learners like GPT-3, Alayrac et al. [54] curated a large-scale multimodal

Table 1
Comparison of different LLM architectures.

Feature	Encoder-Only	Decoder-Only	Encoder-Decoder
Applicable Tasks	Understanding Tasks	Generation Tasks	Sequence-to-sequence Tasks
Computation Complexity	Low	Medium	High
Modeling Direction	Bidirectional	Unidirectional	Unidirectional + Bidirectional
Advantages	Strong semantic understanding, efficient feature extraction	Strong generation capability	Balances understanding and generation
Disadvantages	Cannot generate output sequences, requires fine-tuning	Reliance on autoregressive mechanisms	High complexity, resource-intensive

dataset from the web, comprising primarily text–image pairs, to train an MLLM named Flamingo. Flamingo adapts seamlessly to visual tasks via few-shot learning, eliminating the need for task-specific fine-tuning. Flamingo’s robust multimodal in-context learning (ICL) and few-shot capabilities mark it as the GPT-3 moment for the multimodal domain, positioning it as the beginning of MLLMs. These MLLMs employ LLMs as cognitive engines, preserving their inherent capabilities while integrating powerful visual functionalities [55]. This advancement paves the way for the development of generalist medical AI systems. For example, Med-PaLM M [21] achieved performance levels comparable to or exceeding the state-of-the-art expert models across 14 different medical tasks, showcasing the potential of MLLMs as general medical assistants.

2.5. High-quality data

A key factor behind the success of LLMs and MLLMs is their use of large-scale training data, which enables the acquisition of universal representations transferable to diverse language understanding and generation tasks [38]. However, much of this training data comes from web sources like WebText [47] and Common Crawl, and it is inevitable that there are some toxicities and biases in these large amounts of web data, which are also carried over to LLMs and MLLMs [25]. To mitigate the adverse effects of large-scale datasets and improve model performance, researchers often employ high-quality datasets for fine-tuning.

For example, Li et al. [18] utilized GPT-4 to create open-ended instruction-following data derived from biomedical image-caption datasets, subsequently training LLaVA-Med on this data. LLaVA-Med exhibited remarkable multimodal conversational capabilities, adeptly addressing biomedical image queries based on user instructions. To further refine multimodal medical datasets, Xie et al. [56] introduced a dataset enriched with multi-grained annotations. These annotations encompass global context, modality details, and localized descriptions of medical images. Training LLaVA-Med on this dataset led to a nearly 10% average performance boost across three biomedical VQA datasets, emphasizing the critical role of high-quality datasets. Notably, LIMA [38] fine-tuned using only 1,000 meticulously curated prompts and responses with standard supervised loss, outperformed both Alpaca [57] and Bard in human and GPT-4 preference scores. Ablation experiments on LIMA demonstrated that enhancing data quality provides greater benefits than merely increasing data quantity [38]. Consequently, it can be predicted that data engineering is emerging as a pivotal area of research focus.

This section outlines the development trajectory of LLMs, emphasizing the shift from supervised learning to unsupervised pre-training & fine-tuning, and ultimately to unsupervised pre-training & prompting. The success of LLMs has catalyzed rapid growth in the multimodal domain, leading to the development of MLLMs built on LLM foundations. Notably, GPT-3 and Flamingo, with their robust few-shot capabilities, mark the beginning of LLMs and MLLMs, respectively. Recent studies highlighting the role of high-quality datasets in LLMs and MLLMs suggest that data engineering will emerge as a key research focus. In summary, the evolution of LLMs and MLLMs reflects a progressive shift from initial feature engineering to structure engineering, objective engineering, and now, to prompt and data engineering.

3. Structure of LLMs and MLLMs

Existing LLMs are universally based on the Transformer architecture, which adopts an encoder–decoder framework. Accordingly, these LLMs have evolved into three structural variants based on the Transformer architecture: (1) Encoder-only, represented by models such as BERT [44]; (2) Decoder-only, represented by models such as the GPT series [4,5]; (3) Encoder-Decoder, represented by models like T5 [58]. Table 1 compares the advantages and disadvantages of different LLM architectures. Current MLLMs extend LLMs by integrating a vision encoder to process visual information and a modality alignment module [55,59] to bridge the gap between vision and text modalities. This section provides a comprehensive overview of existing medical LLMs and MLLMs, focusing on their respective model architectures. Section 3.1 reviews medical LLMs categorized by the three aforementioned structures. Section 3.2 discusses the common vision encoders, LLM backbones, and modality alignment techniques employed in medical MLLMs. For clarity, Table 2 and Table 3 detail and categorize existing medical LLMs and MLLMs.

3.1. Structure of LLMs

3.1.1. Encoder-only

Encoder-only language models (LMs) consist of multiple encoder layers within the Transformer architecture, with BERT being the earliest and most representative example. Inspired by BERT, additional encoder-only LMs such as DeBERTa [60], ALBERT [61], and RoBERTa [62] have been developed. Encoder-only LMs commonly utilize the masked language modeling (MLM) task during pre-training, where random tokens in sentences are masked, and the model is trained to predict these tokens accurately. This pre-training approach equips encoder-only LMs with exceptional natural language understanding capabilities, allowing them to effectively encode and comprehend medical knowledge, thereby improving performance in various medical tasks. Consequently, researchers have focused on developing dedicated encoder-only LMs tailored specifically for the medical domain [45,46,63]. For example, BioBERT [46], pre-trained on biomedical corpora, achieved state-of-the-art results in tasks such as biomedical named entity recognition, relation extraction, and QA. MentalBERT [63] was trained on datasets of mental health disorders (e.g., depression, anxiety, suicidal ideation) sourced from social platforms like Reddit and Twitter, facilitating its application in mental health research.

Encoder-only LMs employ a bidirectional attention mechanism, enabling simultaneous modeling of contextual information, which makes them especially well-suited for tasks requiring deep semantic understanding, such as text classification and named entity recognition [64]. Additionally, encoder-only LMs function as robust text embedding extractors, with their semantic representations effectively supporting diverse downstream tasks. Notably, the parallel nature of the MLM task ensures that predictions for each masked token are independent, allowing encoder-only LMs to utilize computational resources efficiently during training, thereby reducing pre-training time significantly [65]. However, encoder-only LMs are designed primarily for input sequence comprehension, lacking generative capabilities, which restricts their use in generation tasks. Furthermore, these models usually require task-specific fine-tuning and lack the strong ICL and few-shot capabilities characteristic of models such as GPT-3. Consequently, encoder-only LMs are commonly categorized as pre-trained language models (PLMs) [33,34] instead of LLMs. Subsequent sections of this survey will not further explore such PLMs.

Table 2

Detailed information on existing medical LLMs categorized by architecture type.

Category	Model name	Base model	Para.(B)	Training data source	Construction method	Evaluation method	Date
Decoder-Only	Med-PaLM [68]	PaLM	540	MultiMedQA	IFT	AEM, Human	2022/12
	ChatDoctor [16]	LLaMA	7	Alpaca-52K, HealthCareMagic-100k	IFT	AI	2023/03
	Baize-Healthcare [69]	LLaMA	7	Quora, MedQuAD	SFT	AI	2023/04
	BenTsao [70]	LLaMA	7	CMeKG	SFT	Human	2023/04
	MedAlpaca [71]	LLaMA	7/13	Medical Meadow	IFT	AEM	2023/04
	PMC-LLaMA [72]	LLaMA	7/13	MedC-K, MedC-I	CPT, IFT	AEM	2023/04
	Med-PaLM 2 [13]	PaLM 2	340	MultiMedQA	IFT	AEM, Human	2023/05
	Clinical Camel [73]	LLaMA 2	13/70	ShareGPT, PubMed, MedQA	SFT	AEM	2023/05
	HuatuoGPT [74]	BLOOMZ	7	Hybrid Data	SFT, RLAI	AEM, Human, AI	2023/05
	GatorTronGPT [75]	GPT-3	5/20	Clinical Text from UF Health, Pile	PT	AEM	2023/06
	ClinicalGPT [76]	BLOOM	7	Three MedQA, MD-EHR, MedDialog	SFT, RLHF	AEM	2023/06
	Zhongjing [77]	Ziya-LLaMA	13	CMtMedQA, ChatMed, CMeKG	CPT, SFT, RLHF	Human, AI	2023/08
	Radiology-Llama2 [78]	LLaMA 2	7	MIMIC-CXR, OpenI	IFT	AEM, Human	2023/08
	MedChatZH [79]	Baichuan	7	Books, med-mix-2M	CPT, IFT	AEM, AI	2023/09
	CPLLM [80]	LLaMA 2	13	eICU-CRD, MIMIC-IV	IFT	AEM	2023/09
	ChatCounselor [23]	Vicuna	7	Psych8k	IFT	AI	2023/09
	Qilin-Med [81]	Baichuan	7	ChiMed	CPT, SFT, DPO	AEM	2023/10
	AlpaCare [82]	LLaMA	7/13	MedInstruct-52k	IFT	AI	2023/10
	TCM-GPT [83]	BLOOM	7	TCM-Corpus-1B, TCM-EXAM, TCM-EHR	CPT, SFT	AEM	2023/11
	HuatuoGPT-II [84]	Baichuan 2	7/13	Web Corpus, Books, Literature, Encyclopedia	IFT	AEM, Human, AI	2023/11
	MEDITRON [85]	LLaMA 2	7/70	GAP-Replay, MedMCQA, PubMedQA, MedQA	CPT, SFT	AEM	2023/11
	AMIE [21]	PaLM 2	340	MedQA, MultiMedBench, MIMIC-III, RealWorld Dialogue	IFT	Human, AI	2024/01
	BioMistral [86]	Mistral	7	PubMed Central	CPT, SFT	AEM	2024/02
	Me-LLaMA [87]	LLaMA 2	13/70	Pile, MIMIC-III, MIMIC-IV, MIMIC-CXR, RedPajama	CPT, IFT	AEM	2024/02
	Apollo [88]	Qwen	7	ApolloCorpora	CPT, SFT	AEM	2024/03
	BioMedLM [89]	Transformer	2.7	PubMed Center, Pile	PT, SFT	AEM	2024/03
	PediatricsGPT [90]	Baichuan 2	7/13	PedCorpus	CPT, SFT, DPO	AEM, Human, AI	2024/05
Encoder-Decoder	DoctorGLM [91]	ChatGLM	6	ChatDoctor, HealthcareMagic, MedDialog, CMD.	IFT	Human	2023/04
	BianQue [92]	ChatGLM	6	BianQueCorpus	IFT	AEM	2023/10
	SoulChat [22]	ChatGLM	6	SoulChatCorpus	IFT	AEM, Human	2023/11

¹ Encoder-only models are not included as they typically belong to the PLM, not LLM.² “CPT” means continuous pre-training, “IFT” means instruction fine-tuning, “SFT” means supervised fine-tuning, “RLHF” means reinforcement learning from human feedback, “RLAI” means reinforcement learning from AI feedback, “DPO” means direct preference optimization.³ “AEM” means automatic evaluation metrics.

3.1.2. Decoder-only

Decoder-only models are the dominant architecture for LLMs, comprising multiple decoder layers within the Transformer. The first decoder-only language model was GPT, and GPT-3 later marked a beginning of LLMs, paving the way for numerous other notable decoder-only models [2,3,5–7]. Decoder-only LLMs primarily use next-token prediction (NTP) as their pre-training objective. In this process, the model learns to predict the next token in a sequence based on all preceding tokens. This training paradigm equips decoder-only LLMs with remarkable generative capabilities, enabling them to convert discriminative tasks into generative ones. This unification of task formats enhances both their generalization and adaptability across application scenarios. For example, Med-PaLM M [21] excels in a range of tasks, such as text-based QA, VQA, image classification, radiology report generation, and summarization, achieving or exceeding state-of-the-art performance.

Compared to encoder-only LMs, decoder-only leverage NTP as the pre-training task, enhancing their capabilities in text generation. These models adopt a unidirectional attention mechanism, which simplifies the training process and facilitates the capture of temporal dependencies in sequence generation. Moreover, with appropriately designed contextual prompts, decoder-only LLMs can reframe numerous NLP tasks as generation tasks, achieving remarkable performance in few-shot and zero-shot learning scenarios [66,67]. This ability is a key factor behind the dominance of the decoder-only structure in current LLM designs. However, unlike encoder-only LMs, the unidirectional attention mechanism of decoder-only LLMs constrains their ability to understand semantics, especially for complex input structures. This limitation often necessitates significantly larger training datasets to mitigate its impact. Furthermore, since each token prediction relies on its predecessor, the parallel processing capacity of decoder-only LLMs is restricted, resulting in substantially longer pre-training durations. This distinction is especially evident when compared to encoder-only LMs [65].

3.1.3. Encoder-decoder

Encoder-decoder LLMs leverage the Transformer architecture, combining stacks of encoders and decoders. The encoder processes input sequence and outputs representations with contextual information, which the decoder uses for text generation. Prominent examples of encoder-decoder LLMs T5 [58] and GLM [118]. Similar to encoder-only and decoder-only architectures, encoder-decoder LLMs have also been adapted for medical applications. For example, Chen et al. [22] fine-tuned ChatGLM using the empathetic dialogue dataset SoulChatCorpus. The resulting model exhibited robust empathetic capabilities, assisting users in articulating their thoughts and offering suitable suggestions during psychological counseling.

Encoder-decoder LLMs integrate the strengths of encoder-only and decoder-only architectures, utilizing the encoder to create rich input text representations and the decoder to generate output sequences, thus balancing comprehension and generative capabilities. These models are especially effective for tasks involving complex input-to-output mappings, such as machine translation and text summarization. However, their complex architecture demands significantly higher computational resources [65]. In comparison to decoder-only models, encoder-decoder LLMs need to simultaneously acquire encoding and decoding capabilities during pre-training, necessitating larger datasets and greater computational resources to perform optimally on certain generative tasks. As demonstrated by Wang et al. [66], decoder-only LLMs outperform in zero-shot scenarios without requiring fine-tuning. By contrast, encoder-decoder LLMs require extensive multi-task fine-tuning on annotated datasets to achieve peak performance. Since the dominant LLM training paradigm is based on unsupervised learning from large corpora, decoder-only architectures, with their superior zero-shot performance, are better equipped to utilize unlabeled data. Consequently, decoder-only architectures continue to dominate as the preferred design for LLMs.

Table 3
Detailed information on existing medical MLLMs.

Modality alignment method	Model name	Vision encoder	LLM backbone	Data source	Evaluation method	Date
GATED XATTN-DENSE Layers	Med-Flamingo [93]	CLIP-ViT	LLaMA	MTB, PMC-OA	AEM, Human	2023/07
	Flamingo-CXR [94]	CLIP-ViT	LLaMA	IND1, MIMIC-CXR	AEM, Human	2024/11
Query-Based	MedBLIP [95]	EVA-CLIP-ViT	BioMedLM	ADNI, NACC, OASIS	AEM	2023/05
	XrayGLM [96]	ViT-G	ChatGLM	MIMIC-CXR, OpenI	/	2023/05
	PCLmed [97]	EVA-CLIP-ViT	ChatGLM	ImageCLEF 2023 caption prediction	AEM	2023/06
	RadFM [98]	3D ViT	MedLLaMA-13B	MedMD, RadMD	AEM, Human	2023/08
	CheXagent [99]	EVA-CLIP-ViT	Mistral	CheXinstruct	AEM, Human	2024/01
Projection-Based	CLIP-ViT w/ GPT2 [100]	CLIP-ViT	GPT2-XL	Slake, PathVQA, OVQA	AEM	2023/05
	MedVInt [101]	PMC-CLIP-ViT	PMC-LLaMA	PMC-VQA	AEM	2023/05
	PathAsst [102]	PathCLIP-ViT	Vicuna	PathCap, PathInstruct	/	2023/05
	LLaVA-Med [18]	CLIP-ViT	LLaMA	PMC-15M, VQA-RAD, SLAKE, PathVQA	AEM, AI	2023/06
	XrayGPT [103]	MedCLIP-ViT	Vicuna	MIMIC-CXR, OpenI	AEM, AI	2023/06
	Med-PaLM M [21]	ViT-e, ViT-22B	PaLM	MultiMedBench	AEM, Human	2023/07
	R2GenGPT [20]	Swin Transformer	LLaMA 2	IU-Xray, MIMIC-CXR	AEM	2023/09
	Qilin-Med-VL [104]	ViT	LLaMA-2-Chinese	ChiMed-VL	/	2023/10
	MAIRA-1 [105]	RAD-DINO	Vicuna	MIMIC-CXR	AEM, Human	2023/11
	PeFoM-Med [106]	EVA-CLIP-ViT	LLaMA 2	ROCO, VQA-RAD	AEM, Human	2024/01
	M3D-LaMed [107]	3D ViT	LLaMA-2	M3D-Data	AEM, AI	2024/03
	MoE-TinyMed [108]	CLIP-ViT	Phi-2	LLaVA-Med, VQA-RAD, SLAKE, PathVQA	AEM	2024/04
	MAIRA-2 [109]	Rad-DINO	Vicuna	MIMIC-CXR, PadChest, USMix	AEM	2024/06
	PathChat [110]	UNI	LLaMA 2	PubMed, WSIs	AEM, Human	2024/06
	HuatuoGPT-Vision [111]	CLIP-ViT	Yi-1.5	PubMedVision, HuatuoGPT-II	AEM	2024/06
	miniGPT-Med [112]	EVA-CLIP-ViT	LLaMA 2	MIMIC, NLST, SLAKE, RSNA, RadVQA	AEM	2024/07
	SkinGPT-4 [113]	ViT	LLaMA 2	SKINCON, Dermnet	Human	2024/07
	LLaVA-Med++ [56]	CLIP-ViT	LLaMA	MedTrinity-25M, VQA-RAD, SLAKE, PathVQA	AEM	2024/08
	SigPhi-Med [114]	SigPhi	Phi-2	LLaVA-Med, VQA-RAD, SLAKE, PathVQA	AEM	2024/10
Prompt Augmentation	ChatCAD [17]	Expert models	ChatGPT	MIMIC-CXR, CheXpert	AEM	2023/02
	Visual Med-Alpaca [115]	Expert models	Med-Alpaca	ROCO, BigBio	/	2023/04
	ChatCAD+ [116]	Expert models	ChatGPT	CheXpert, MIMIC-CXR	AEM	2023/05
	OphGLM [117]	Expert models	ChatGLM	Web data, MedDialog	AEM	2023/06

3.2. Structure of MLLMs

As shown in Fig. 4, this section provides a detailed discussion of three critical modules in MLLMs: the Vision Encoder, the LLM Backbone, and the Modality Alignment Module. The method of leveraging expert models to construct MLLMs is treated as a form of prompt augmentation method [115] and is discussed alongside other modality alignment modules.

3.2.1. Vision encoder

MLLMs extend LLMs by incorporating a vision encoder, thereby equipping them with visual capabilities. Specifically, the vision encoder V encodes visual input I_x into visual features Z_x , illustrated below:

$$Z_x = V(I_x) \quad (2)$$

Various options exist for the vision encoder V , such as ResNet [119], the ViT [120], and CLIP-ViT [52]. Here, we provide several insights regarding different vision encoders. (1) Transformer-based vision encoders typically outperform CNN-based counterparts due to their superior scalability. Pre-training on large-scale datasets enables these encoders to extract more effective visual representations. Furthermore, as most LLMs are based on the Transformer architecture, Transformer-based vision encoders integrate more seamlessly with these models. (2) Chen et al. [121] observed that contrastive-learning-based vision encoders often outperform classification-based ones, particularly in localization and visual-text understanding tasks. This advantage likely arises from the pre-training tasks of contrastive-learning-based encoders, which inherently involve vision-language alignment, enabling the extraction of visual features more aligned with semantic spaces for MLLMs. (3) In the medical field, vision encoders pre-trained on medical imaging datasets consistently outperform those trained on natural scene datasets. For example, Li et al. [18] demonstrated in their LLaVA-Med experiments that employing BioMed CLIP-ViT [122] as the vision encoder significantly enhanced LLaVA-Med's performance across three medical VQA benchmarks. This enhancement is likely due to significant differences in color, texture, and structure between medical images

and natural scene images. Vision encoders pre-trained on medical data are better adapted to the unique distributions of medical images, thereby achieving superior performance on related tasks. Furthermore, evidence from models such as MAIRA-1 [105] and PathChat [110] reinforces this perspective. Consequently, in the medical domain, optimizing the vision encoder, rather than merely scaling up the model, is regarded as a more cost-effective strategy, especially under constrained computational resources.

In summary, while ResNet remains a strong choice among CNNs, Transformer-based ViT models are increasingly preferred by researchers. Additionally, contrastive-learning-based ViT models, such as CLIP-ViT and EVA-CLIP ViT [123], often outperform classification-pretrained ViT models when used as vision encoders for MLLMs. Consequently, ViT models trained using contrastive learning represent the mainstream choice for vision encoders. Furthermore, vision encoders pre-trained on medical imaging datasets significantly enhance the performance of medical MLLMs.

3.2.2. LLM backbone

As the cognitive engine of MLLMs, the LLM backbone is the most critical module among the three primary components and contains the largest number of parameters. It equips MLLMs with capabilities such as text interaction, ICL, and reasoning. The operational principle of the LLM backbone in MLLMs is illustrated below:

$$R = L(H_x, T_x) \quad (3)$$

Where L represents the LLM backbone, R denotes the response output of the LLM, T_x indicates the embedded tokens of the text input, and H_x are visual representations that LLM can understand. The specific meaning of H_x is explained in Eq. (4).

Although powerful LLMs like ChatGPT have not yet been open-sourced, numerous high-quality open-source LLMs are available for researchers. Among these, the LLaMA series [6–8] developed by Meta, stands out as one of the most popular open-source LLMs and is frequently used as the backbone for MLLMs. Additionally, fine-tuned versions of LLaMA, such as Vicuna-13B [124], achieve performance

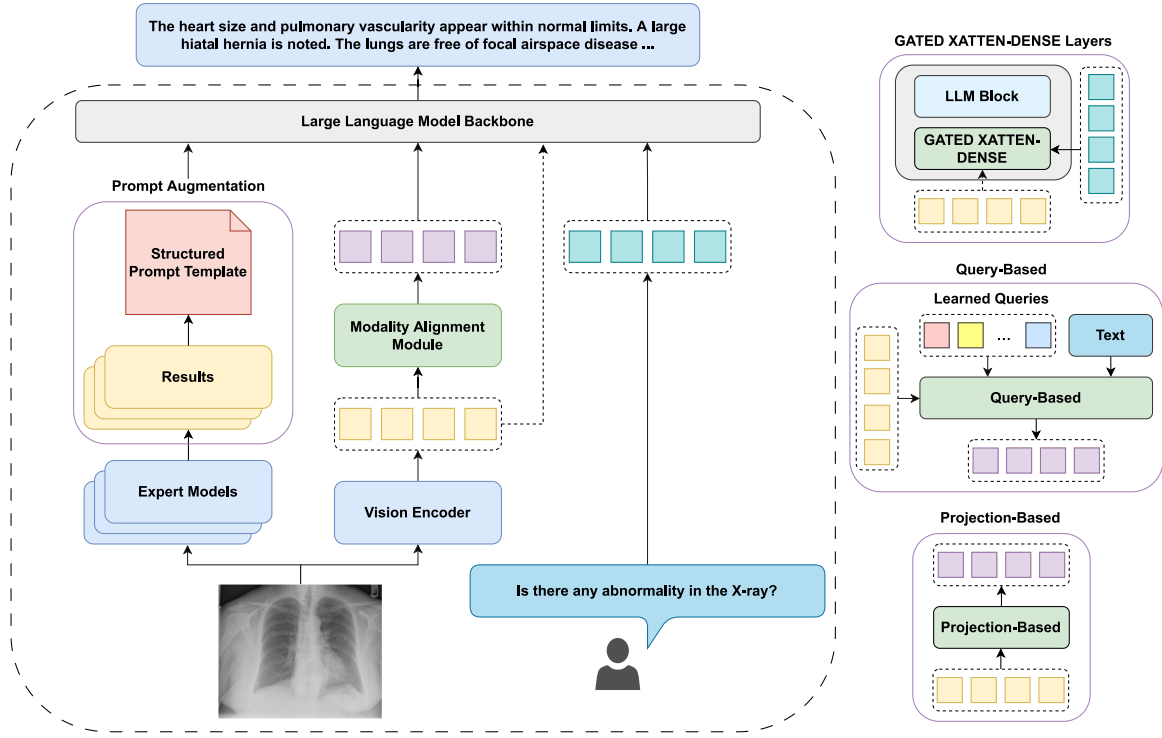


Fig. 4. The core modules and pipeline of MLLMs. On the far right are three types of modality alignment modules. The approach of utilizing expert models to construct MLLMs is regarded as a type of prompt augmentation method, classified under modality alignment modules for further elaboration.

comparable to 90% of ChatGPT and Bard. Notably, different models demonstrate varying levels of performance across languages. For example, Mistral [125] excels in French, Qwen [126] is optimized for Chinese, and GPT-4 offers robust support for a broader range of languages. Consequently, researchers can choose LLM backbones based on specific linguistic requirements.

3.2.3. Modality alignment

Although integrating a vision encoder into LLMs enables them to process visual inputs, LLMs trained exclusively on text datasets cannot interpret the output features Z_x produced by the vision encoder's output. Consequently, modality alignment is required to transform Z_x into a format that LLMs can understand, as shown in Eq. (4):

$$H_x = f(Z_x) \quad (4)$$

Where f represents the modality alignment method, and H_x refers to visual representations that LLMs can understand. Modality alignment plays a critical role in enabling MLLMs to interpret visual information and significantly enhances their multimodal capabilities. The subsequent sections introduce four established modality alignment methods: GATED XATTN-DENSE Layers, Query-Based method, Projection-Based method, and Prompt Augmentation.

GATED XATTN-DENSE Layers, introduced in Flamingo [54], incorporate dense cross-attention layers into a frozen pre-trained LLM. These cross-attention layers receive information from the vision encoder's output, which is typically processed through a Perceiver Resampler [127] to reduce the computational complexity of vision-text cross-attention. Using additional cross-attention layers, the LLM generates text responses based on visual representations. Subsequent works such as Med-Flamingo [93], which are based on Flamingo, also utilize these cross-attention layers for modality alignment.

Query-Based method, often considered a multimodal perceiver [128], extracts information from visual representations through a set of learnable query vectors. For example, the Q-Former introduced in BLIP-2 [129] extracts visual features from a frozen vision encoder, enabling LLMs to generate text responses aligned with visual information. This

query-based approach can be effectively extended to 3D spaces, as demonstrated by Chen et al. [95] with MedBLIP, which adapts the querying mechanism for 3D medical imaging. Although these methods can represent images with only a few queries, thus reducing training costs, they risk losing critical visual information. Moreover, Yao et al. [130] demonstrated that the Q-Former functions merely as an inefficient visual token compressor. For cost-effective token reduction, adaptive average pooling outperforms it.

Projection-Based method can be regarded as a type of multi-modal converter [128]. It is simpler than the query-based method, as it maps visual representations from the vision encoder's output to the word embedding space using a simple projection layer, allowing LLMs to interpret images. For example, LLaVA-Med, Qilin-Med-VL, and XrayGPT [103] employ a simple linear layer to map visual representations, and MedVIntTE [101] and LLaVA-1.5 [131] rely on MLPs for this purpose. The mapped visual representations, combined with textual representations, act as inputs to the LLM backbone. Compared to query-based methods, projection-based methods preserve more visual information because they do not reduce the number of vision tokens, though this advantage comes at the cost of increased training complexity.

Prompt Augmentation typically processes images using expert models, integrates their results with text prompt templates to serve as input for LLMs, thereby linking visual information with text. For example, OphGLM [117] extracts information from fundus images through classification and segmentation models, embeds this information into structured text templates, and forms diagnostic reports that act as inputs to LLMs. Similarly, in ChatCAD [17], X-ray images are processed by trained computer-aided diagnosis (CAD) models to generate outputs, which are subsequently transformed into natural language via prompt templates and used as inputs to LLMs. Compared to query-based and projection-based methods, prompt augmentation leverages expert models without requiring additional modality alignment training. However, this method heavily depends on the quality of prompt templates and the performance of expert models.

Overall, the GATED XATTN-DENSE Layers method integrates visual and textual features by incorporating cross-attention layers preceding each layer of the LLMs [54]. However, compared to the other three modality alignment methods, this approach demands substantially higher computational resources. For example, training the Flamingo model utilized 1,535 TPuv4 devices and 828.9k GPU hours, vastly exceeding the 128 GPU hours required for LLaVA. Moreover, studies have shown that projection-based and query-based models outperform those using GATED XATTN-DENSE Layers on VQA benchmark tasks [59,132]. Consequently, models utilizing GATED XATTN-DENSE Layers may necessitate a more extensive hyperparameter search process to achieve comparable performance. The query-based method, designed to extract the most relevant information from visual features for specific tasks, serves as a strategy for compressing visual information. For example, in BLIP-2 [129], the Q-Former compresses image features into 32 tokens, substantially reducing computational costs. However, Q-Former employs a lossy compression approach, potentially leading to the loss of spatial positional information in images [133]. Furthermore, compared to the projection-based method, Q-Former demonstrates lower training efficiency and underperforms simpler pooling strategies [130,134]. The prompt augmentation approach avoids additional modality alignment training but depends on fixed prompt templates, constraining its flexibility. Furthermore, translating images into textual descriptions risks information loss, potentially resulting in hallucinations in MLLMs. In contrast, the projection-based method remains the most widely used and straightforward modality alignment approach. It transfers visual information to LLMs without loss via a projection layer, ensuring minimal information loss. Additionally, compared to the query-based method, the projection-based approach features fewer parameters, faster convergence, and superior scalability. Despite their differences, all four approaches share a text-centered design philosophy, utilizing text as a modality space to transform visual information into textual space, thus enabling LLMs to process visual input.

4. Principles of medical LLMs and MLLMs

To assist researchers and medical professionals in developing their own medical LLMs and MLLMs, this section summarizes the medical datasets available for training, explains the methods for fine-tuning medical LLMs and MLLMs, and finally discusses three approaches for evaluating the performance of medical LLMs and MLLMs.

4.1. Training datasets

The currently available datasets are categorized into six main types: electronic health records (EHRs), scientific literature, QA, dialogue, medical image-text pairs, and instruction-following data. Table 4 provides detailed information about these datasets.

Electronic Health Records: EHRs include personal health records, such as demographic information, summaries of major diseases and health issues, and primary healthcare records. The Medical Information Mart for Intensive Care III (MIMIC-III) [135] is one of the largest and most widely used publicly available EHR datasets, containing approximately 2 million de-identified notes across 13 specialties, including cardiology, respiratory medicine, and radiology. The MIMIC-III dataset is a valuable resource for developing medical LLMs, as evidenced by models such as AMIE [159] and GatorTron [160], which were trained using this dataset. Besides MIMIC-III, other widely used EHR datasets include the Clinical Practice Research Datalink (CPRD) [137] and the updated version of MIMIC-III, MIMIC-IV [136].

Scientific Literature: Scientific literature, which provides accurate and authoritative medical knowledge, serves as a key source for medical datasets. PubMed, the most widely used repository for biomedical and life science literature, provides access to key resources, including MEDLINE, PubMed Central (PMC), and the NCBI Bookshelf. It indexes citations from over 34 million biomedical research articles. PubMed

abstracts include approximately 4.5 billion words, making it an excellent resource for medical training datasets. In addition to PubMed, PubMed Central (PMC) is a widely used repository offering free full-text access, with its articles collectively containing around 13.5 billion words. PubMed and PMC offer high-quality medical literature, often used as sources for other datasets. For instance, PMC-OA [157], PMC-VAQ [101], and PMC-15M [158] are three biomedical multimodal datasets extracted from PMC, significantly facilitating the development of medical LLMs [72,73,85] and MLLMs [18,93].

Question-Answer: QA datasets are categorized into two types: discriminative QA [14,140] and generative QA [101]. Discriminative QA datasets primarily include multiple-choice questions, whereas generative QA focuses on open-ended questions. Typical QA datasets include PubMedQA [139], MedQA [14], PMC-VQA [101], and Multi-MedQA [68]. MultiMedQA, in particular, is a comprehensive medical QA dataset encompassing seven sub-datasets that assess the authenticity, helpfulness, accuracy, and potential harm of LLMs' responses. Beyond text-based QA datasets, the medical domain also includes VQA datasets. For example, the classic VQA-RAD [148] and SLAKE [149] are prominent VQA datasets comprising radiology images, questions, and answers, which span various organs and anatomical regions. Such medical VQA datasets have greatly contributed to the advancement of medical MLLMs.

Dialogue: While datasets like EHRs, scientific literature, and QA pairs enrich LLMs and MLLMs with medical knowledge, relying exclusively on these may result in insufficient long-dialogue interaction capabilities, limiting their practical clinical application. As a result, researchers are focused on developing high-quality dialogue datasets to enhance the models' capabilities in multi-turn conversations. For example, Li et al. [16] curated HealthCareMagic-100k by collecting around 100 K authentic doctor-patient dialogues from the Health-CareMagic platform, followed by extensive filtering. To bypass the labor-intensive process of gathering authentic dialogue datasets – requiring extensive filtering and deduplication – Li et al. [16] simulated real dialogue scenarios using ChatGPT to create a synthetic dataset named GenMedGPT-5k.

Image-Text Pairs: In addition to the aforementioned VQA datasets, image-text pairs, including image-caption and image-report datasets, are essential for training medical MLLMs. These datasets offer crucial context and annotations, enriching the ability of MLLMs to interpret visual information in medical scenarios. For example, PMC-OA [157] comprises 1.65 million medical image-text pairs sourced from PMC, and it has been utilized to train models like PMC-CLIP [157] and Med-Flamingo [93]. Zhang et al. [101] built upon PMC-OA and utilized ChatGPT to generate a diverse set of high-quality QA pairs. After filtering, they created PMC-VQA, which provides 227 K VQA pairs. PMC-15M [158], derived from PMC articles, includes 15 million figure-caption pairs, making it two orders of magnitude larger than MIMIC-CXR [151]. Other notable medical image-text datasets, such as ChiMed-VL [104], RadMD [98], and Open-I [152], have also played a significant role in advancing medical MLLMs.

Instruction-Following Data: The effectiveness of medical LLMs and MLLMs in downstream tasks depends not only on the medical knowledge they acquire but also on their capacity to follow user instructions. Enhanced instruction-following ability enable more accurate comprehension and execution of user directives, thereby improving downstream task performance. To strengthen the instruction-following ability of medical LLMs and MLLMs, training on instruction-following datasets has become widely accepted. These datasets often include instruction-response pairs or image-instruction-response, such as PathInstruct [102] with 180 K image-instruction-response pairs, LLaVA-Med-Instruct [18] with 60 K image-instruction-response pairs, and ChiMed-VL-Instruction [104] with 469 K image-instruction-response pairs. Such datasets effectively enhance models' instruction-following ability, leading to improved zero-shot performance. Notably, instructions can include explicit commands or questions; as a result,

Table 4
Summary of medical datasets for pre-training and fine-tuning.

Datasets	Type	Description	AI synthesis
<i>Datasets For Medical LLMs</i>			
MIMIC-III [135]	EHR	Approximately 2M de-identified notes.	✗
MIMIC-IV [136]	EHR	About 300K patients, 430K admissions.	✗
CPRD [137]	EHR	Anonymized medical records for over 11.3M patients.	✗
PubMed	Literature	Over 34M citations and abstracts of biomedical literature, about 4.5B words.	✗
PMC	Literature	Provides free full-text access to PubMed, about 13.5B words.	✗
CORD-19 [138]	Literature	More than 140K papers, with more than 72K full text.	✗
PubMedQA [139]	QA	1K labeled, 612K unlabeled and 211.3K manually generated QA.	✗
MedQA (USMLE) [14]	QA	61,097 multiple-choice QA pairs.	✗
MedMCQA [140]	QA	194K multiple-choice QA pairs.	✗
cMedQA2 [141]	QA	100K questions and 200k answers.	✗
MultiMedQA [68]	QA	Includes six existing datasets and one new dataset.	✗
MedQuAD [142]	QA	47,457 question-answer pairs from trusted medical sources.	✗
Medical Meadow [71]	QA	Over 160K QA pairs.	✓
Huatuo-26M [143]	QA	26M QA pairs.	✗
Psych8k [23]	QA	8,187 query-answer pairs.	✓
CMtMedQA [77]	Dialogue	70K multi-round conversation datasets from real doctor-patient conversations.	✓
MedDialog [144]	Dialogue	3.4M Chinese conversations and 0.6 million English conversations.	✗
HealthCareMagic-100k [16]	Dialogue	100K authentic patient-doctor conversations.	✗
GenMedGPT-5k [16]	Dialogue	5K generated conversations between patients and physicians from ChatGPT.	✓
MedC-I [72]	Instruction-Following Data	202M tokens.	✓
MedInstruct-52k [82]	Instruction-Following Data	52K instruction-response pairs generated by GPT-4.	✓
UMLS [145]	Knowledge Base	2M entities for 900K concepts.	✗
CMeKG [146]	Knowledge Base	Chinese medical knowledge graph.	✗
COMETA [147]	Web Data	Consisting of 20K English biomedical entity mentions.	✗
TCM-Corpus-1B [83]	Web Data	20GB dataset collected from Baidu Baike, Wikipedia and other sources.	✗
ChiMed [81]	Hybrid	Composed of various data such as QA, books, dialogues, etc.	✗
GAP-REPLAY [85]	Hybrid	Includes data from clinical practice guidelines, abstracts, and original articles.	✗
<i>Datasets For Medical MLLMs</i>			
PMC-VQA [101]	QA	Contains 149K images, 227K VQA pairs.	✓
VQA-RAD [148]	QA	315 radiology images and 3515 QA pairs generated by clinicians.	✗
Slake [149]	QA	642 radiology images and over 7000 diverse QA pairs.	✗
PathVQA [150]	QA	4,998 pathology images with 32,799 QA pairs.	✗
MIMIC-CXR [151]	Image-Report	227,835 imaging studies for 65,379 patients.	✗
OpenI [152]	Image-Report	7,470 images and 3,955 reports.	✗
CheXpert [153]	Image-Report	224,316 chest X-rays with reports.	✗
ROCO [154]	Image-Caption	Contains more than 81K radiologic images, each with a corresponding title, keywords.	✗
OpenPath [155]	Image-Caption	208,414 pathology images paired with natural language descriptions.	✗
MediCaT [156]	Image-Caption	160K images with captions and inline references.	✗
PathCap [102]	Image-Caption	142K high quality pathology image-caption pairs.	✓
MedMD [98]	Image-Caption	15.5M 2D scans, 180k 3D scans, with corresponding captions or diagnosis labels.	✓
PMC-OA [157]	Image-Caption	1.6M image-caption pairs.	✗
PMC-15M [158]	Image-Caption	15M figure-caption pairs from over 3M articles.	✗
LLaVA-Med-Alignment [18]	Image-Caption	600K image-caption pairs from PMC-15M.	✓
ChiMed-VL-Alignment [104]	Image-Caption	580,014 images and context information or descriptions.	✓
PubMedVision-Alignment [111]	Image-Caption	647,031 image-caption pairs.	✓
MedTrinity-25M [56]	Image-Annotation	Contains 25M samples along with their multigranular annotations.	✓
LLaVA-Med-Instruct [18]	Instruction-Following Data	60K instruction-following data.	✓
ChiMed-VL-Instruction [104]	Instruction-Following Data	469,441 question-answer pairs.	✓
PubMedVision-Alignment [111]	Instruction-Following Data	647,031 instruction-following data.	✓
PathInstruct [102]	Instruction-Following Data	180K instruction-following data.	✓
CheXInstruct [99]	Instruction-Following Data	An instruction-tuning dataset curated from 28 publicly available datasets	✓
ApolloCorpora [88]	Hybrid	2.5B tokens with data in multiple languages.	✗
PedCorpus [90]	Hybrid	The corpus includes pediatric textbooks, clinical guidelines, and knowledge graphs.	✓
M3D-Data [107]	Hybrid	Comprising 120K image-text pairs and 662K instruction-response pairs.	✓

¹ "Hybrid" means that the dataset is a mixture of multiple types of data.

² "AI Synthesis" indicates that generative AI such as chatGPT and GPT-4 were used during the development of the dataset to assist in generating the data.

some works [18,104] also considered QA data as instruction-following data.

In addition to the aforementioned data types, various medical knowledge bases [145] and web data [83] also serve as training sources for medical LLMs and MLLMs. Different types of data are typically utilized at various training stages. For medical LLMs, scientific literatures and web data are primarily used during pre-training and continual pre-training phases to incorporate medical knowledge and facilitate domain adaptation. In contrast, QA, dialogue, and instruction-following data are typically employed during the fine-tuning phase to enhance interaction capabilities and instruction-following performance. For medical MLLMs, image-caption pairs are widely used during pre-training to align visual features with text representations, while

instruction-following data is commonly applied for fine-tuning. Furthermore, the extensive content of EHRs and scientific literature often makes them foundational sources for other data types. For example, PMC-15M [158] is derived from 3 million PMC articles, yielding 15 million image-caption pairs. Finally, studies have demonstrated that fine-tuning models with substantial high-quality synthetic data generated by ChatGPT can significantly enhance downstream task performance [161]. As a result, AI-assisted data generation has emerged as a prevalent strategy in the data-scarce medical field. For example, Liu et al. [23] transformed 260 real psychological counseling audio recordings into text and utilized GPT-4 to extract question-answer pairs and generate key summaries, providing supplementary contextual information for constructing the Psych8k dataset.

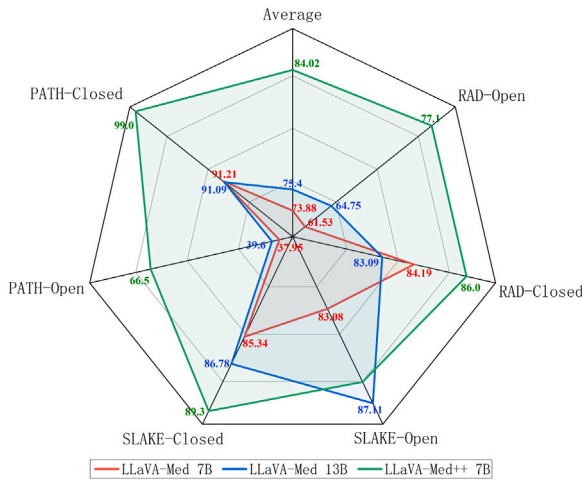


Fig. 5. Performance of LLaVA-Med 7B, LLaVA-Med 13B, and LLaVA-Med++ 7B on three medical VQA benchmarks. Although LLaVA-Med 13B has a larger number of parameters, the performance improvement is minimal. In contrast, LLaVA-Med++ 7B, trained on a larger and higher-quality dataset under the same settings, shows a significant improvement in performance on the benchmarks.

Although increasing model size often results in substantial performance gains in general domains due to abundant training data, the scarcity of medical data has emerged as a critical bottleneck, hindering the performance improvements of medical LLMs and MLLMs. Even with larger model sizes, performance gains are not consistently substantial. For example, experiments conducted by Li et al. [18] revealed that scaling LLaVA-Med from 7B to 13B yielded negligible performance improvements (as shown in Fig. 5). In contrast, under identical conditions, LLaVA-Med++ [56] trained on a larger and higher-quality dataset like MedTrinity-25M [56], exhibited significant performance improvements on medical VQA benchmark tasks. Consequently, in the medical domain, the development and exploration of high-quality medical datasets represent a pivotal strategy for improving model performance, while solely increasing model size proves to be an inefficient approach.

4.2. Fine-tuning methods

The extensive parameters in LLMs and MLLMs make training medical LLMs and MLLMs from scratch computationally intensive. Consequently, the prevalent method for constructing medical LLMs and MLLMs involves fine-tuning general foundation models using medical datasets. This section outlines six fine-tuning methods, as illustrated in Fig. 6, to aid researchers in developing medical LLMs and MLLMs. In addition, this section explores the characteristics of these fine-tuning methods, providing practical guidance for selecting the appropriate fine-tuning strategies.

Continual Pre-Training: CPT [34,162] refers to the further pre-training of a general pre-trained model on large-scale medical datasets. Given that general LLMs and MLLMs often lack domain-specific medical knowledge, CPT aims to incorporate such knowledge into these models. For LLMs, datasets comprising extensive medical texts, including scientific literature, books, and clinical guidelines, are typically used in CPT to ensure LLMs acquire sufficient medical knowledge. For example, MEDITRON-70B [85], which is based on LLaMA 2, utilizes a medical mixed dataset composed of clinical guidelines, PubMed papers, and abstracts for CPT. For MLLMs, medical image-caption pairs are frequently utilized as CPT datasets. This method aligns visual features with the semantic space by training MLLMs to predict captions for medical images. Prominent models, including LLaVA-Med [18], Qilin-Med-VL [104], and HuatuoGPT-Vision [111], utilize large-scale medical image-caption datasets for CPT.

Instruction Fine-Tuning: While CPT using large-scale biomedical corpora integrates medical knowledge into LLMs and MLLMs, clinical performance also relies on their instruction-following capability. A lack of instruction-following ability in LLMs and MLLMs can result in unpredictable behavior, underscoring the necessity of fine-tuning these models with instruction-following datasets to improve their compliance with diverse human instructions [18,163]. Instruction fine-tuning (IFT) enables models to accurately understand and execute human directives, thereby significantly enhancing their zero-shot performance. For example, Singhal et al. [68] demonstrated substantial improvements in Flan-PaLM's performance on MedQA, MedMCQA, and PubMedQA after fine-tuning it with instruction data.

Supervised Fine-Tuning: To further improve the performance of LLMs and MLLMs in downstream tasks, SFT is typically performed on datasets tailored to those tasks. For instance, Chen et al. [85] fine-tuned MEDITRON on MedQA, PubMedQA, and MedMCQA datasets to enhance its performance in medical QA tasks. Similarly, Li et al. [18] improved LLaVA-Med's performance in medical VQA tasks by fine-tuning it on PathVQA, SLAKE, and VQA-RAD datasets. Hyland et al. [105] leveraged large-scale image-report pairs to enhance MAIRA-1's performance in radiology report generation tasks. While prior studies often do not strictly distinguish between SFT and IFT, we argue that IFT focuses on utilizing instruction datasets to enhance models' instruction-following ability and zero-shot performance. Conversely, SFT targets high-quality task-specific datasets to fine-tune LLMs and MLLMs, thereby boosting their performance in specialized downstream tasks.

Reinforcement Learning from Human Feedback: Reinforcement learning from human feedback (RLHF) [5,164] is a technique for aligning model behavior more closely with human preferences and directives. Compared to the preceding three fine-tuning approaches, RLHF is more intricate and involves three distinct stages [5,164,165]: collecting human feedback, training the reward model, and policy optimization, as shown in Fig. 7. During the human feedback collection stage, the primary task is gathering comparison data. Typically, an LLM is provided with a prompt, generating multiple outputs that experts annotate and score based on quality [165]. These annotated outputs, along with the prompts, form the comparison data. For example, Yang et al. [77] employed 6 medical graduate students or clinical doctors as labelers to rank the model's outputs based on dimensions such as safety, professionalism, and fluency, forming a comparison dataset. During reward model training, a reward model learns from the comparative data to produce scalar rewards that represent human preferences. During policy optimization, a new prompt is provided as input to the LLM, whose response is evaluated by the reward model, outputting a scalar reward. The LLM is then fine-tuned using Proximal Policy Optimization (PPO) based on these rewards. Notably, the reward model's data quality is typically lower than that used for SFT [33], directly transitioning from pre-training to RLHF may lead to suboptimal fine-tuning results, so RLHF is often conducted after IFT and SFT [7,164].

Reinforcement Learning from AI Feedback: Reinforcement learning from AI feedback (RLAIF) is a cost-effective alternative to RLHF, where the reward model learns from AI feedback without requiring human annotation [166]. In the medical domain, Zhang et al. [74] sampled multiple responses from the fine-tuned model after IFT and SFT and used ChatGPT to evaluate them across dimensions such as informativeness, coherence, adherence to human preferences, and accuracy. This comparison data was then used to train a reward model. Training reward models through AI feedback eliminates the need for manual data labeling in RLHF, significantly reducing labor costs.

Direct Preference Optimization: While RLHF and RLAIF align models with human preferences and ethical norms, they typically involve fitting a reward model that reflects human preferences and combining reinforcement learning to fine-tune LLMs and MLLMs. However, this process is both complex and often unstable. Direct preference optimization (DPO) [167] is a simpler and more efficient paradigm for aligning models with human preferences, bypassing the need for a

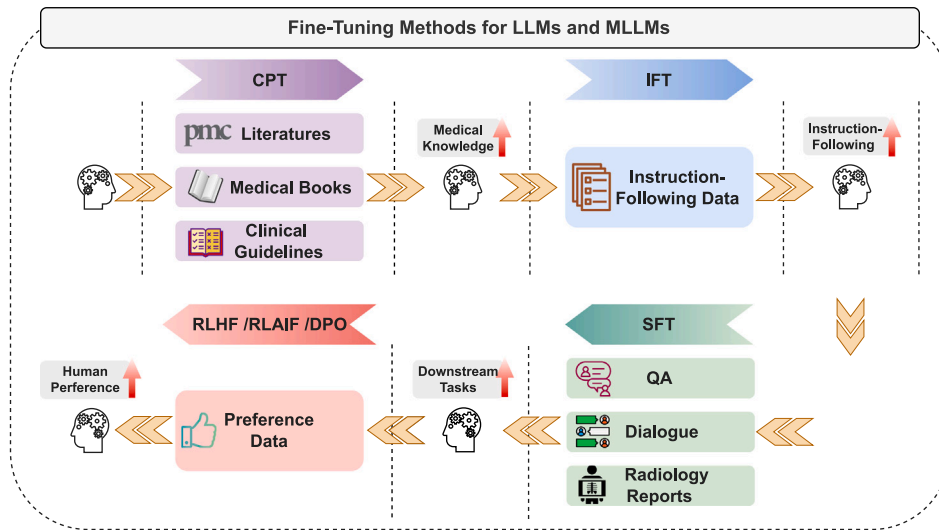


Fig. 6. Overview of six fine-tuning methods. In our analysis of the related work on medical LLMs and MLLMs, we found that Continuous Pre-Training (CPT) is commonly used to inject medical knowledge into LLMs and MLLMs; Instruction Fine-Tuning (IFT) enhances the models' ability to follow instructions and their zero-shot performance; Supervised Fine-Tuning is frequently employed to improve model performance on specific tasks; and Reinforcement Learning from Human Feedback (RLHF), Reinforcement Learning from AI Feedback (RLAIF), and Direct Preference Optimization (DPO) are used to align model behavior with human preferences.

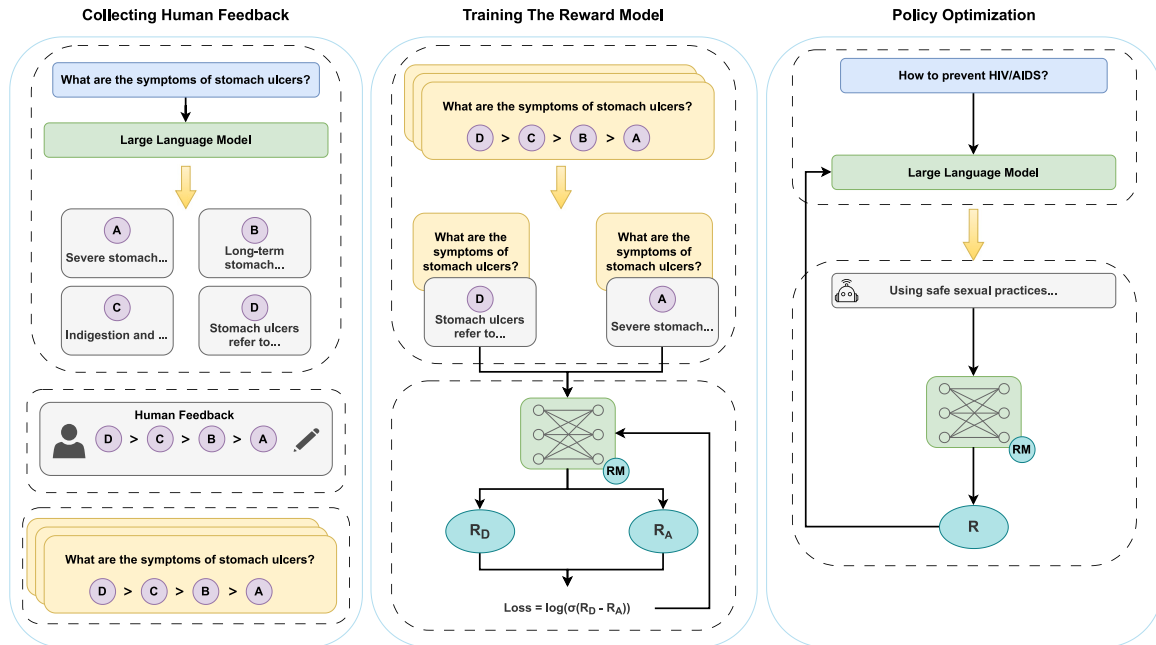


Fig. 7. Pipeline of Reinforcement Learning from Human Feedback. **Left** illustrates the Collect Human Feedback phase: A labeler provides a single prompt to the model, ranks multiple responses, and collect the prompt along with the labeled responses. **Mid** shows the Reward Model Training phase: A prompt and two responses are randomly sampled from the dataset and used to train the reward model. **Right** represents the Policy Optimization phase: A new prompt is provided, and the reward model generates a scalar reward for the model's response, which is then used for policy optimization.

reward model by directly optimizing the model using preference data. The core idea of DPO is to use an analytical mapping from the reward function to the optimal policy, transforming the reward function loss into a policy loss and thereby eliminating the need for explicit reward modeling. For example, Qilin-Med [81] uses two publicly available preference datasets after SFT to optimize the model through DPO, achieving stable and efficient training while aligning it with human preferences.

Overall, LLMs and MLLMs can initially acquire medical terminology, common knowledge, and related concepts through CPT on large-scale medical corpora. However, training on large-scale corpora incurs significant computational overhead. Additionally, medical datasets for CPT typically originate from scientific literature, textbooks, and clinical

guidelines but lack conversational examples from real clinical settings, limiting LLMs and MLLMs in developing conversational and question-answering abilities for practical clinical use. Further training is required to enhance interaction capabilities. IFT and SFT are widely used methods for fine-tuning medical LLMs and MLLMs. IFT mainly enhances the model's instruction-following ability and zero-shot performance with instruction-following data, whereas SFT aims to improve performance on specific downstream tasks. However, the training objectives of both methods differ from the actual generative task objectives, potentially resulting in exposure bias [168,169]. Specifically, during inference, LLMs and MLLMs generate text by: (1) predicting the next token based on the existing text; (2) appending the predicted token to the existing text; and (3) repeating this process until a stop token is generated.

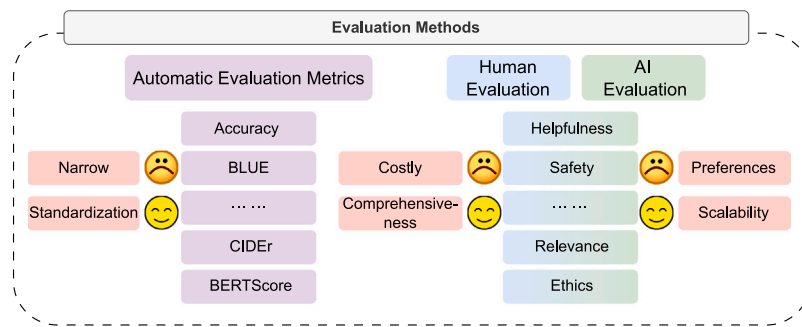


Fig. 8. Overview of three evaluation methods. We summarize three methods for evaluating medical LLMs and MLLMs: Automatic Evaluation Metrics, Human Evaluation, and AI Evaluation, and discuss their respective advantages and disadvantages.

During training, however, the token appended in Step 2 is typically the ground truth token rather than the model's predicted token. This discrepancy causes the loss function to overestimate the model's generative capabilities. RLHF and RLAI are effective in mitigating exposure bias. In these methods, LLMs and MLLMs autonomously generate a complete response based on the input instructions, instead of incrementally relying on ground truth tokens to predict the next token. Subsequently, the reward model evaluates the quality of the generated response. However, the training objectives of RLHF and RLAI differ significantly from those of IFT and SFT, often resulting in instability during the training process [5]. Although DPO addresses certain stability issues, it similarly depends on constructing comparison datasets. In the medical domain, this process demands substantial input from medical experts, significantly increasing the cost of data construction.

4.3. Evaluation methods

With the continuous emergence of abilities in medical LLMs and MLLMs, coupled with growing ethical and safety concerns [170], comprehensively evaluating their performance in medical tasks and ensuring their safety in clinical environments has become a pressing challenge. Therefore, this subsection summarizes three commonly used evaluation methods and examines their respective advantages and limitations, as shown in Fig. 8.

4.3.1. Automatic evaluation metrics

Accuracy is commonly used to evaluate the performance of medical LLMs and MLLMs on multiple-choice question benchmarks, such as MedQA [14] and MedMCQA [140]. However, accuracy alone is insufficient for evaluating tasks requiring longer text generation, such as medical report writing and summarization. Consequently, the evaluation of medical LLMs and MLLMs must incorporate additional metrics for a more comprehensive assessment.

Bilingual Evaluation Understudy (BLEU) [171] metric assesses the quality of generated text by measuring the similarity of n -grams (consecutive word sequences of length n) between the generated and reference texts. BLEU is categorized into BLEU-1, BLEU-2, BLEU-3, and BLEU-4 based on the value of n , capturing n -gram similarity at different levels. For instance, BLEU-1 reflects word-level accuracy, while BLEU-4 emphasizes text continuity. Recall-Oriented Understudy for Gisting Evaluation (ROUGE) [172] includes metrics such as ROUGE-N, ROUGE-L, ROUGE-W, and ROUGE-S. Similar to BLEU, ROUGE-N evaluates n -gram similarity between generated and reference texts; however, it emphasizes recall, whereas BLEU focuses on precision. ROUGE-L assesses textual coherence by calculating the longest common subsequence between the generated and reference texts. ROUGE-W extends ROUGE-L by applying weighted scoring, giving higher importance to continuous and accurate matching subsequences. ROUGE-S extends ROUGE-N by accommodating non-contiguous words in n -grams. Google BLEU (GLEU) [173], a BLEU variant, incorporates lexical overlap and

word order, offering a more nuanced evaluation of fluency and naturalness. The Distinct- n [174] metric quantifies text diversity by calculating the ratio of unique n -grams to total n -grams. CIDEr [175], tailored for image caption evaluation, combines n -gram recall and precision, assigning higher weights to rare n -grams to assess key information. BERTScore [176] leverages pre-trained BERT embeddings to compute token-level similarity scores between generated and reference sentences. Compared to n -gram-based metrics, BERTScore more effectively evaluates vocabulary and compositional diversity.

In the medical domain, models such as HuaTuoGPT [74], Clinical-GPT [76,76], SoulChat [22] and BianQue [92] commonly use these metrics to assess generative performance. While these metrics partially capture the accuracy and fluency of generated text, they fall short in assessing clinical dialogue quality [159] and alignment with human values, necessitating the inclusion of human evaluation.

4.3.2. Human evaluation

Human evaluation is an essential approach to assess medical LLMs and MLLMs, as it addresses aspects that automatic evaluation metrics may fail to capture. For example, Tu et al. [159] highlighted that metrics such as BLEU and ROUGE fail to reflect the clinical quality of medical consultations. To address this, 23 medical experts from the United States, the United Kingdom, and India were invited to evaluate model-generated responses based on accuracy, appropriateness, and comprehensiveness. Similarly, Yang et al. [77] engaged human experts to assess the safety, accuracy, and ethical implications of model-generated responses.

Clearly, human evaluation can encompass critical aspects such as safety and helpfulness, which are essential for medical LLMs and MLLMs. Despite its ability to evaluate diverse capabilities of medical LLMs and MLLMs, human evaluation remains inherently subjective due to the absence of standardized criteria. Moreover, hiring medical experts is costly, making AI evaluation a practical alternative.

4.3.3. AI evaluation

Employing advanced AI models, such as ChatGPT and GPT-4, which align with human values, has become the predominant method for evaluating medical LLMs and MLLMs [177]. Wang et al. [177] conducted experiments on five natural language generation datasets, showing that ChatGPT, as an evaluation tool, outperformed traditional metrics in most cases and matched human evaluation. In the medical domain, Li et al. [18] asked GPT-4 to evaluate responses from itself and LLaVA-Med on criteria such as helpfulness, relevance, accuracy, and level of detail. Liu et al. [23] instructed GPT-4 to evaluate whether LLM responses were acceptable and whether their tone resembled that of human counselors.

Despite its scalability and reduced reliance on human input, AI evaluation has notable limitations. Studies [69,178] have revealed that GPT-4, as an evaluation tool, tends to favor the first response when multiple answers are presented sequentially. Furthermore, GPT-4 tends

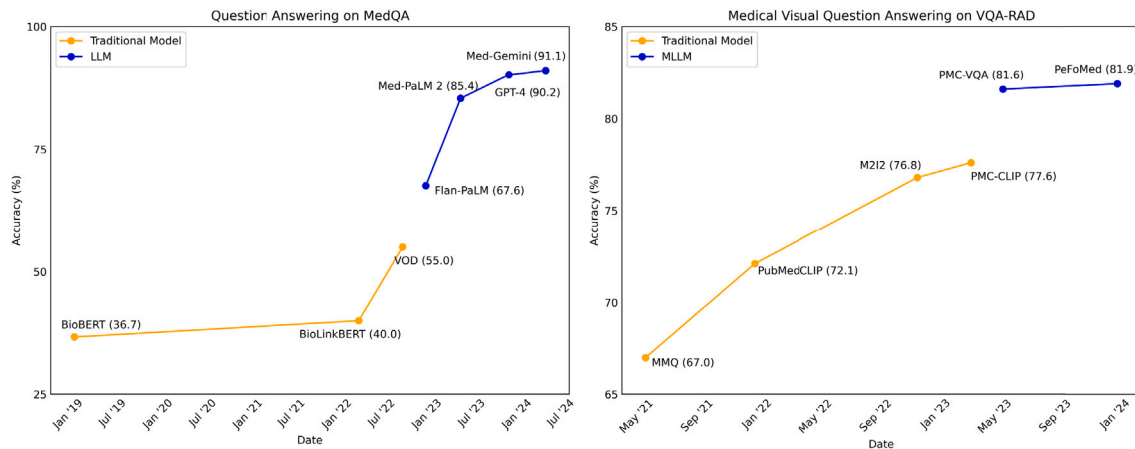


Fig. 9. Performance comparison between traditional deep learning models and medical LLMs/MLLMs on QA and VQA benchmark datasets. Recent advancements in medical LLMs/MLLMs demonstrate their significant superiority over traditional deep learning models on QA and VQA benchmark datasets.

to prefer longer responses and those it has generated itself [23]. To mitigate the limitations of the aforementioned methods, integrating multiple evaluation approaches may provide more reliable results. Additionally, training specialized LLMs or MLLMs through reinforcement learning or other methods to align with human judgment criteria could overcome AI evaluation's shortcomings.

5. Applications of LLMs and MLLMs in medicine

Traditional medical models are tailored for specific tasks, including medical named entity recognition, relation extraction, text classification, and semantic textual similarity. Although these models perform well in their respective tasks, they are limited in their ability to integrate multi-source data for complex clinical applications. In contrast, medical LLMs and MLLMs excel in navigating diverse medical contexts and performing a wide range of tasks, while also gradually surpassing traditional deep learning models in single-task performance, as shown in Fig. 9. Consequently, medical LLMs and MLLMs act as versatile medical assistants with extensive application potential. To aid practitioners in comprehending the developmental trajectory of LLMs and MLLMs in medicine, this section highlights their potential applications in healthcare (Fig. 10) and outlines strategies for employing these models in diverse medical tasks.

5.1. Medical diagnosis

AI has been under development in medical diagnosis for several decades [179]. While some breakthroughs have been achieved, its role has largely been confined to assisting with some simple diagnostic tasks, such as medical image segmentation, lesion detection and classification. In recent years, with advancements in LLMs and MLLMs, doctors and patients are now expected to rely on these models for more comprehensive diagnosis. Specifically, these models can process subjective descriptions of disease symptoms or medical images, such as X-rays, and leverage embedded medical knowledge to directly generate diagnostic results. For physicians, LLMs and MLLMs can serve as practical supporters, ensuring faster and more effective care, while also reducing diagnostic times and workloads [180]. For patients, they offers detailed information about their condition along with recovery suggestions.

Currently, Med-PaLM 2, one of the top-performing medical LLMs, generates responses to consumer medical and adversarial questions, outperforming physician-generated answers across multiple evaluation criteria [13]. This underscores the potential of LLMs as medical diagnostic assistants. Additionally, Yuan et al. [181] showed that multi-turn dialogues with GPT-4 significantly improve its ability to accurately

diagnose and recommend effective treatments for gastrointestinal cancers, achieving a performance level comparable to experienced physicians. In prostate cancer, Zhu et al. [182] developed 22 questions based on patient education guidelines and clinical experience, addressing topics like screening, prevention, treatment options, and postoperative complications. Testing indicated that ChatGPT provided accurate and comprehensive responses, while demonstrating appropriate humanistic care toward patients. Yang et al. [83] further advanced the application of LLMs as diagnostic assistants by training TCM-GPT on traditional Chinese medicine datasets, showing that it outperformed other models in traditional Chinese medicine examination and diagnosis, contributing to the advancement of traditional Chinese medicine. In particular, a study by Goh et al. [183] explored whether LLMs could improve doctors' diagnostic reasoning abilities. The study recruited 50 doctors with 2 to 8 years of clinical experience and randomly divided them into two groups: one group used GPT-4 as an auxiliary tool in addition to traditional diagnostic resources, while the other group relied exclusively on traditional resources. Both groups diagnosed six challenging clinical cases, with the experimental results presented in Table 5. For each case, the GPT-4-assisted group achieved a median diagnostic reasoning score of 76%, compared to 74% for the group relying solely on traditional resources. Additionally, doctors using GPT-4 had a median diagnostic time of 519 s per case, while those relying solely on traditional resources took 565 s. With GPT-4's assistance, doctors demonstrated modest improvements in both diagnostic scores and time. Notably, GPT-4 achieved a diagnostic score of 92% when independently diagnosing the six clinical cases, significantly outperforming any group of clinicians, regardless of its use as an auxiliary tool. This phenomenon may stem from doctors' general lack of trust in the diagnostic results provided by LLMs or their insufficient training in systematically integrating LLMs as diagnostic tools in clinical practice.

Furthermore, inspired by general MLLMs [10,11], researchers have developed multimodal medical diagnostic assistants [17,18,21,96,103,115,184], expanding diagnostic capabilities from purely text-based input to incorporating medical images. For example, Zhou et al. [113] introduced SkinGPT-4, a model that autonomously analyzes images, identifies skin condition features and categories, conducts in-depth analysis, and offers interactive treatment recommendations. To evaluate the reliability and robustness of SkinGPT-4, Zhou et al. [113] carried out a comprehensive study involving 150 real-world cases, where the model's diagnostic outcomes were compared with dermatologists' assessments. The study results revealed a substantial agreement between the diagnoses provided by SkinGPT-4 and those made by dermatologists. In 150 cases, dermatologists rated the model's diagnoses as correct or relevant in 80.63% of the cases. Furthermore, SkinGPT-4 demonstrated rapid response times, analyzing each case in just a few

Table 5
Diagnostic analysis of six clinical cases by different groups.

Group	Median diagnostic score	Median diagnostic time per case
Doctors Using GPT-4	76%	519 s
Doctors Not Using GPT-4	74%	565 s
GPT-4 Independent Diagnosis	92%	N/A

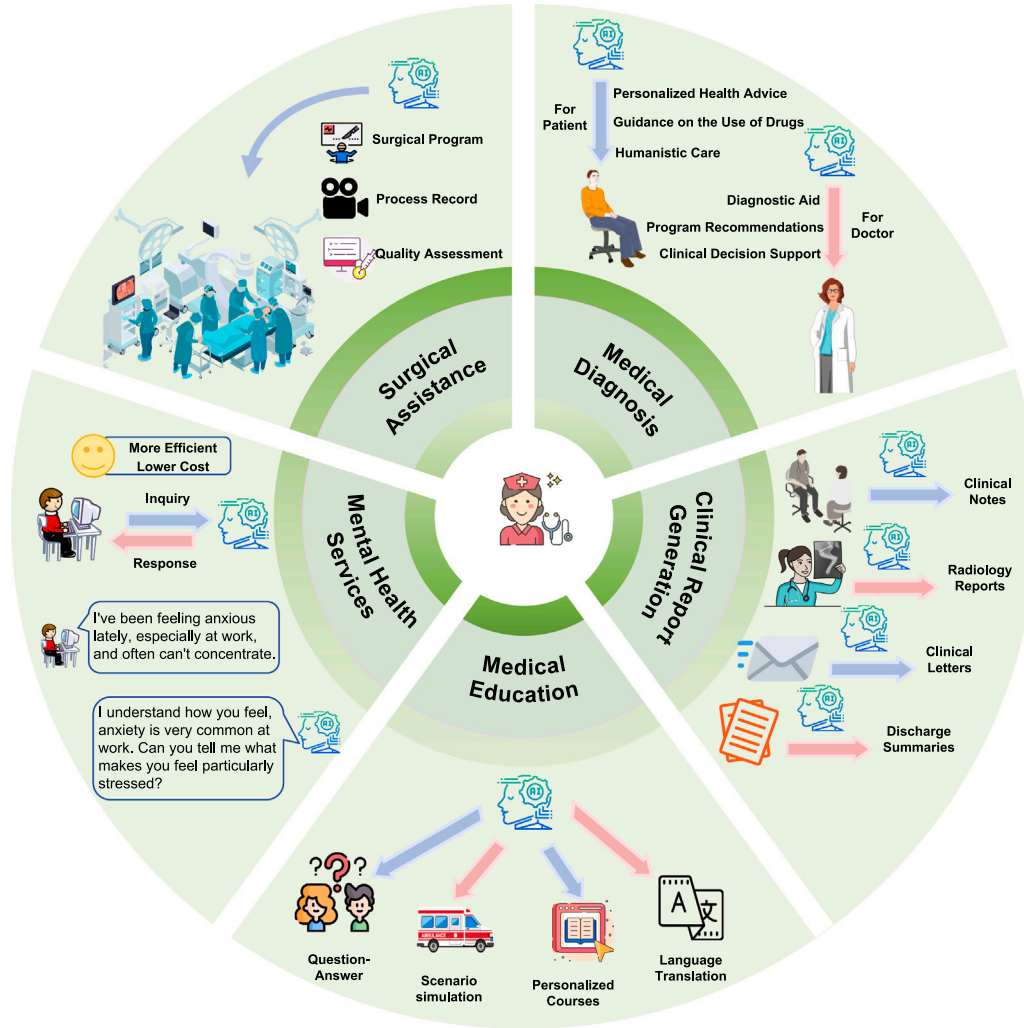


Fig. 10. Overview of potential applications of LLMs and MLLMs in medicine.

seconds, whereas dermatologists took an average of over 10 min. To further validate the consistency of SkinGPT-4's diagnostic outcomes, the research team randomly selected 45 cases, each of which was independently diagnosed ten times. The consistency of diagnoses for the same case reached 93.73%, further confirming the reliability and effectiveness of SkinGPT-4 as a diagnostic tool in dermatology.

Medical LLMs and MLLMs, as medical diagnostic assistants, are capable of providing diagnostic recommendations for doctors and offering consultation advice for patients. However, due to inherent limitations of LLMs and MLLMs [29,170], these models currently serve only as auxiliary tools for physicians, with their diagnostic outputs considered as references rather than definitive results.

5.2. Clinical report generation

Clinical reports are standardized documents prepared by doctors for patients. The manual drafting of clinical reports is often tedious

and time-consuming, significantly increasing clinicians' workload and reducing overall efficiency. Medical LLMs and MLLMs, equipped with extensive medical knowledge and generative capabilities, serve as efficient tools for clinical report generation.

For example, during medical consultations, doctors typically record key information from patient interactions, which serves as a basis for evaluating conditions or informing other clinical reports. Medical LLMs can function as clinical note-taking tools, automating this task for doctors [73]. Doctors can provide LLMs with doctor-patient interaction records, which the models process to generate detailed medical notes [185]. Doctors can also prompt LLMs to simplify medical notes by removing complex details and generating concise summaries for easier review and analysis [19].

Following a medical diagnosis, doctors often draft diagnostic documents such as radiology reports. Medical MLLMs, capable of processing visual inputs, are particularly effective in assisting with radiology report generation. For example, miniGPT-Med, developed by Alkhaldi

et al. [112], achieved state-of-the-art performance in generating medical reports, surpassing prior models in accuracy by 19%. Additionally, Tanno et al. [94] proposed Flamingo-CXR, an MLLM specifically designed for generating chest X-ray (CXR) reports. To evaluate the clinical quality of Flamingo-CXR-generated reports, the research team conducted a pairwise preference test with 27 radiologists. In this test, radiologists compared Flamingo-CXR-generated reports with those written by human radiologists, selecting their preferred option. Results indicated that on the MIMIC-CXR dataset, Flamingo-CXR-generated reports were rated as equivalent to or superior to original clinical reports in 56.1% of cases. This proportion increased to 77.7% on the IND1 dataset [186], and further to 94% for normal cases within the IND1 dataset. Moreover, the study demonstrated that radiologists' report quality improved when supported by Flamingo-CXR. On the MIMIC-CXR dataset, 53.6% of reports authored by radiologists with Flamingo-CXR's assistance were rated as equivalent to or superior to the original clinical reports. This figure increased to 71.2% on the IND1 dataset. These findings underscore the feasibility of employing medical MLLMs for radiology report generation.

During treatment, doctors explain the cause of the disease, the treatment process, and provide detailed clinical information to patients through clinic letters. By utilizing LLMs to generate clinic letters, clinicians can streamline this tedious process, with the resulting letters exhibiting coherence, accuracy, and empathy comparable to those created by humans [187]. After patient recovery, clinicians allocate significant time to drafting discharge summaries, potentially delaying patient discharge. By employing LLMs, clinicians can generate complete discharge summaries in seconds by providing a template and necessary inputs [188]. The quality of these summaries often surpasses those produced by junior doctors [189].

Leveraging advanced LLMs and MLLMs, various clinical reports from patient admission to discharge can be automatically generated. These reports are more comprehensive and accurate than those produced by humans [19,189], significantly alleviating doctors' workloads and allowing them to dedicate more time to patient care [188]. However, these LLMs and MLLMs are intended to serve exclusively as auxiliary tools for generating clinical reports. They can draft, modify, and summarize reports, but the final versions must be reviewed, edited, and approved by clinicians, who remain accountable for their content [12,25].

5.3. Medical education

GPT-4 and Med-PaLM 2 passed the USMLE with scores exceeding 86% [190], while GPT-4V [191] achieved 90.7%, outperforming most medical students on medical image-related questions [192]. This demonstrates that certain LLMs and MLLMs possess the capability to provide educational support for medical students. Moreover, LLMs and MLLMs can role-play in various contexts, enabling the simulation of diverse learning scenarios for users. Consequently, platforms like Khanmigo [193] and Duolingo [194] have integrated tools like GPT-4 to enhance online teaching and learning.

Specifically, medical LLMs and MLLMs can simulate patients in scenarios such as accidents, emergency rooms, or operating rooms, offering simulation training to medical students prior to clinical practice to enhance their professional and practical skills—tasks unachievable by traditional deep learning [195]. For example, Yamamoto et al. [196] developed an LLM-based simulated patient system and investigated its potential to improve medical students' interviewing skills in a controlled trial. A total of 145 students participated in the study, with 35 assigned to the experimental group, where they trained by interacting with simulated patients, and the remaining 110 forming the control group and receiving traditional education. The main evaluation criterion was the score on the Pre-Clinical Clerkship Objective Structured Clinical Examination (pre-CC OSCE), a standardized test assessing medical interviewing skills. Results revealed that the experimental group

scored an average of 28.1 on the pre-CC OSCE, significantly outperforming the control group's average score of 27.1. Furthermore, survey responses suggested that students found the AI-simulated patients to be highly realistic and capable of facilitating strong emotional interactions [196]. Students also highlighted the flexibility and convenience of this training method, which enabled repeated practice and increased their confidence. Beyond enhancing interview skills, LLMs acting as simulated patients offered medical students structured feedback on their patient history-taking dialogues. Holderried et al. [197] examined 106 interactions between medical students and GPT-4, functioning as simulated patients. The analysis revealed that over 99% of the feedback generated by GPT-4 was considered reasonable and exhibited near-perfect consistency with evaluations by human experts. Additionally, medical LLMs and MLLMs can evaluate students' performance in simulated exercises and create personalized learning plans, a process that is typically time-intensive for teachers but can be executed more cost-effectively and efficiently using these models [198]. Finally, given their extensive training on large corpora, medical LLMs and MLLMs excel at translation, offering cross-language capabilities as well as the ability to simplify medical terminology into plain language [199], greatly assisting medical students in reading and writing. In summary, powerful LLMs and MLLMs can enrich medical students' learning experiences by offering comprehensive medical content, personalized curricula, and realistic, diverse scenarios, thereby broadening their horizons in the medical field and laying a strong foundation for clinical practice.

The potential of LLMs and MLLMs in medical education surpasses that of traditional training courses, as educators in these courses cannot always interact with students or offer tailored learning plans. While these models hold great promise in medical education, they should be regarded solely as auxiliary tools and not as replacements for educators, as their inherent biases and hallucinations make it challenging for students to evaluate the accuracy of the generated content [200,201]. If these models frequently deliver inaccurate content that is hard to identify, they risk misleading students over time. Thus, LLMs and MLLMs can only serve a supportive role in medical education, requiring students to use them under the guidance and supervision of educators.

5.4. Mental health services

Growing societal pressures have led to an increased global demand for mental health services, yet many regions face a severe shortage of mental health specialists due to limited development and resources [202]. Conversation-driven psychological counseling is a central aspect of mental health services, making chatbots powered by LLMs a potential solution for delivering such services in the future [22,23].

For example, Zhong et al. [203] demonstrated that AI chatbot interventions can substantially alleviate patients' depression and anxiety symptoms in the short term, with optimal results often observed approximately eight weeks after the intervention starts. Moreover, compared to professional mental health experts, LLM-based mental health chatbots are more accessible and can extend mental health services to remote or underserved areas lacking specialists. Furthermore, LLM-based chatbots can offer personalized interaction styles tailored to patients' histories and interaction records, including specific emotional patterns, styles, or tones [204]. Additionally, the high cost of psychological counseling and therapy often deters individuals from seeking mental health services. However, LLM-based chatbots can substantially lower costs [15,205], making these services more accessible. Finally, research indicates that individuals are more inclined to share negative emotions with chatbots, as some topics may feel uncomfortable to discuss with humans but easier to disclose to a robot [206]. For example, Grosshans et al. [207] reported a case in which a patient faced difficulties establishing a therapeutic relationship with a psychologist. The patient often felt ashamed during discussions about her condition and discontinued therapy after only a few sessions, despite consulting three different therapists. However, the patient reported

that interacting with an anonymous chatbot enabled her to openly express her fears and suppressed emotions without shame, significantly alleviating her psychological distress. Moreover, the patient highlighted the chatbot's 24/7 availability and immediate responses as particularly crucial in managing her condition. In terms of convenience, cost, and acceptability, LLM-based mental health chatbots outperform mental health professionals, potentially encouraging more individuals with mental health conditions to seek help [204].

Given that patients with mental illnesses are often more vulnerable and psychologically sensitive, mental health service chatbots must exhibit empathy, trustworthiness, understanding, and comfort in conversations, beyond merely offering advice [22]. While research is advancing the empathetic capabilities of LLMs [22], they still fall short compared to humans. Moreover, despite efforts to align LLMs with human concepts and ethical norms using approaches like SFT and RLHF, they may still produce aggressive or psychologically harmful content [170], which is unacceptable for vulnerable mental health patients. Before deploying LLMs as mental health chatbots in real-world applications, significant work is required to address these challenges, along with stricter control measures for such products.

5.5. Surgical assistance

Medical robots have undergone rapid development over the past few decades, significantly enhancing surgeons' capabilities [208] and expanding the potential for minimally invasive surgery [209]. Recently, medical robots have entered a new phase with the advent of MLLMs, which not only endow them with visual capabilities but also enhance interactivity and create a more user-friendly environment.

Efforts are currently underway to explore the application of MLLMs in surgical procedures [210]. Integrating MLLMs into surgical robots allows them to perform essential auxiliary tasks, such as assisting with endoscopic examinations [25]. For instance, the CARES Copilot¹ developed by InnoHK has been tested in the neurosurgery department at Prince of Wales Hospital. During endoscopic surgeries, the system analyzes the procedure in real time and provides the surgeon with prompts, such as information on the current stage of the operation and key considerations for subsequent steps. Additionally, the robust visual capabilities and specialized knowledge of MLLMs enable them to generate valuable diagnostic insights and propose feasible surgical solutions based on endoscopic images. Moreover, during surgical procedures, MLLMs can integrate video streams to annotate the surgical process, analyze and summarize the steps taken, and document non-compliant operations, aiding surgeons in post-surgical reviews.

While medical MLLMs exhibit promising potential for surgical assistance and may play a role in specific medical scenarios, they are not yet suitable for emergency surgeries. This is due to the fact that erroneous information from MLLMs could adversely impact the surgeon's judgment, potentially leading to irreversible consequences. Additionally, current research on MLLMs predominantly focuses on vision-text modalities. We anticipate future investigations will explore other modalities, such as audio and time series, to enable surgical robots to perform more comprehensive and accurate auxiliary tasks while providing more flexible interaction methods.

6. Challenges of LLMs and MLLMs in medicine

Although LLMs and MLLMs have generated significant interest in the AI community and achieved initial successes in medicine, the unique characteristics of the medical field present numerous challenges and risks for their development and deployment, as shown in Fig. 11. In this section, we will discuss and analyze the current challenges faced by LLMs and MLLMs in the medical field in detail, as well as propose possible solutions to these challenges.

6.1. Hallucination phenomenon

Hallucinations refer to the generation of seemingly plausible yet unverified or incorrect information by LLMs and MLLMs [29,30]. This can result in issues such as misleading radiology reports and the dissemination of incorrect medical knowledge in medical education [185]. Therefore, addressing the hallucination problem in LLMs and MLLMs is crucial for accelerating the application of these technologies in medicine.

To tackle this challenge, some researchers have proposed a new benchmark dataset specifically for testing hallucinations in medical LLMs and MLLMs [28]. However, these benchmark datasets can only detect hallucination phenomena in models and do not directly mitigate the problem. Other studies indicate that LLMs primarily acquire knowledge during the pre-training phase [38], and the presence of noisy data, such as error messages in the training dataset, can contribute to hallucinations. Thus, the most fundamental approach to reducing hallucinations is to manually or automatically clean unreliable data from the pre-training corpus [30]. However, the pre-training corpus of LLMs and MLLMs typically consists of vast amounts of data, including web-crawled information, which is challenging to clean and necessitates effective selection and filtering strategies. Consequently, fine-tuning LLMs and MLLMs with high-quality medical datasets is an effective strategy for mitigating hallucinations [211,212]. To further reduce the cost of addressing hallucinations, existing efforts have focused on solutions during the inference stage. For instance, prompting LLMs or MLLMs to verify their own responses has proven effective in alleviating hallucinations [185]. One such method, Chain-of-Verification (CoVe) [213], involves the model drafting an initial response, planning verification questions based on that response, answering these questions to validate the draft, and ultimately generating an optimized answer. Experiments have demonstrated that self-verification methods, such as CoVe, can effectively reduce hallucinations across various tasks. Additionally, retrieval-augmented generation has proven effective in reducing hallucinations [214]. This approach allows the model to retrieve relevant knowledge from external webpages or knowledge bases during the response generation phase [16,102], significantly addressing the hallucination problem.

Particularly, for MLLMs, the occurrence of hallucinations may be attributed to limited visual perception capabilities. For example, the Q-former uses only 32 learnable vectors to represent an image, which inevitably leads to a loss of visual information. Furthermore, the resolution of vision encoders in current medical MLLMs typically ranges from 224 to 336, hindering their ability to capture the complexities of biomedical images and contributing to hallucinations. To tackle this challenge, integrating more advanced and higher-resolution vision encoders, such as DINOv2 [215] and SAM [216], into medical MLLMs represents a viable solution [217].

6.2. Training and deployment challenges

Due to the need for real-time data processing and enhanced privacy protection, local deployment of private LLMs and MLLMs is essential, particularly in medical settings. However, the substantial increase in parameters for LLMs and MLLMs significantly escalates the demand for computational resources, resulting in high operational costs. For example, MEDITRON-70B requires 128 A100 GPUs for training, and the smaller LLaVA-Med 7B necessitates 8 A100 GPUs. Furthermore, even after training and fine-tuning, the large sizes of medical LLMs and MLLMs result in high deployment and inference costs, presenting considerable challenges for most hospitals in locally deploying these models and applying them in practical settings. To facilitate the training and deployment of medical LLMs and MLLMs in hospitals with limited computational resources, this subsection proposes three solutions: optimizing the training process, reducing model parameters, and modifying model architectures.

¹ <https://www.cares-copilot.com/>

To achieve efficient training of LLMs and MLLMs while minimizing computational overhead, researchers have proposed a series of parameter-efficient fine-tuning methods [218–220]. These approaches involve freezing the majority of parameters in LLMs and MLLMs and updating only a small subset to enable effective training. For example, Hu et al. [219] introduced LoRA, which freezes the pre-trained model weights and injects trainable rank decomposition matrices into each layer of the Transformer architecture to facilitate efficient training. However, while PEFT methods facilitate efficient training, they do not address the challenges associated with deployment. To address this issue, reducing model parameters and designing lightweight models are viable solutions [114,221,222]. For example, MobileVLM [221] is a customized MLLM for mobile scenarios, which reduces the training and inference budget by downsizing LLaMA and designing an efficient projector. It is capable of running on mobile devices while remaining competitive with other MLLMs across most tasks.

Currently, most LLMs and MLLMs are based on the Transformer architecture, which leads to a quadratic increase in computational complexity with sequence length, resulting in low efficiency for long sequences. To fundamentally address the challenges of training and deploying medical LLMs and MLLMs, selecting model architectures that are more efficient in computation and inference is a viable option [223, 224]. For example, RWKV [223] combines the efficient parallel training of Transformers with effective inference from RNNs, ensuring constant computational and memory complexity during inference while maintaining comparable performance to similarly scaled Transformer models. Furthermore, Mamba [224], based on the State Space Model, outperforms Transformer models in both performance and inference speed, achieving five times the inference speed of Transformers while remaining comparable in scale. Extending these computationally and inference-efficient model architectures to medical LLMs and MLLMs will help address the current training and deployment challenges faced by these models.

6.3. Lack of recency

Once medical LLMs and MLLMs are trained, the knowledge they acquire becomes static. However, as medical knowledge is continuously updated, the absence of new concepts will exacerbate the models' inaccuracies and hallucination problems. This is particularly evident when the models encounter new terms that are not present in the training corpus, rendering them unable to comprehend this knowledge [12]. For example, if medical LLMs and MLLMs are trained exclusively on data prior to 2020, they will lack information regarding COVID-19. This limitation may prevent the models from understanding terms such as "COVID-19" or "Long COVID," or they may incorrectly classify COVID-19 as a known viral variant (e.g., SARS or MERS), leading to the provision of misleading advice. Consequently, the lack of recent knowledge will significantly hinder the real-world application of medical LLMs and MLLMs.

To address the lack of recency resulting from the offline learning of medical LLMs and MLLMs, continual parameter updates through fine-tuning methods to synchronize them with current human knowledge is a feasible solution [162]. While fine-tuning can inject new medical concepts and knowledge into the model, it also introduces two challenges: catastrophic forgetting, in which the model forgets previously learned knowledge upon acquiring new information [225]. The second challenge is negative forward transfer, wherein performance on unseen tasks deteriorates when learning new tasks [226]. To address these issues, researchers have proposed model editing [227], one is introducing additional trainable parameters to correct erroneous responses stemming from outdated knowledge while preserving the original parameters of the model [228,229]. Another approach involves identifying parameters related to specific knowledge and updating them accordingly to integrate relevant new information [230–232]. In addition to model editing, retrieval-augmented generation can be

employed to update the knowledge of medical LLMs and MLLMs by linking the model to an information retrieval component. This allows the model to retrieve relevant content from external knowledge bases as Refs. [16,102], thereby generating more reliable responses.

6.4. Privacy and security

Medical LLMs and MLLMs are trained on a large-scale medical corpus that includes data such as EHRs, doctor–patient dialogues, and other information that may involve patient privacy, including names, phone numbers, and email addresses. This information can potentially be directly extracted from medical LLMs or MLLMs using specific prompting methods [233,234], raising significant privacy and security concerns. It is crucial to address all privacy and security concerns associated with medical LLMs and MLLMs, ensuring that patient privacy is maintained during the collection and processing of training data [180].

Currently, a common practice to enhance patient privacy protection is data de-identification [16,23]. This process involves removing or anonymizing sensitive information from datasets, including names, phone numbers, addresses, and medical record numbers. Additionally, sensitive terms in clinician–patient dialogues and medical records are replaced or removed to dissociate these terms from specific patients. Moreover, differential privacy methods can effectively mitigate the risk of privacy breaches by adding noise to obscure individual information in the training data, thus preventing the inference of specific details while still enabling meaningful data analysis [235]. Furthermore, utilizing high-quality synthetic data generated by models such as ChatGPT or GPT-4 during training ensures both the controllability and diversity of the datasets while mitigating the risk of privacy leaks. It is also advisable to monitor and filter the model's outputs. For instance, if the output contains sensitive data such as names or contact information, this information should be removed or modified during post-processing. Finally, we urge developers to adhere to ethical standards, ensuring that models respect privacy rights when handling patient data and provide patients with the right to access and delete their information.

6.5. Bias and toxicity

Large-scale corpora, particularly data sourced from the internet, inevitably contain various biased viewpoints, which LLMs and MLLMs may learn [26,236], including biases related to race [237], gender [238], and politics [239]. Additionally, language models may generate toxic responses, such as aggressive and hurtful remarks, with certain groups being more likely to be targeted due to these biases [170]. These biases and toxicities extend to LLMs and MLLMs, posing potential implications and threats to patients, and may have serious consequences for individuals with mental illness.

Reducing bias in training data is a fundamental approach to mitigating bias in models. Specifically, careful curation and screening of diverse, balanced, and representative training data ensure that models learn from a broader range of perspectives and experiences, leading to a more comprehensive understanding and reduced biases across various dimensions [236]. For addressing model toxicity, utilizing empathetic data has been shown to decrease the output of toxic content from models [240]. However, re-screening pre-training datasets and retraining models to minimize biases and toxicities can be costly. Therefore, screening high-quality datasets with anti-bias and anti-toxicity for fine-tuning is a more cost-effective approach. In addition to training, further enhancement is needed in evaluating model bias and toxicity. Designing a comprehensive benchmark for evaluating model bias and toxicity facilitates the detection of these issues, allowing developers to conduct regular reviews of the models [236].

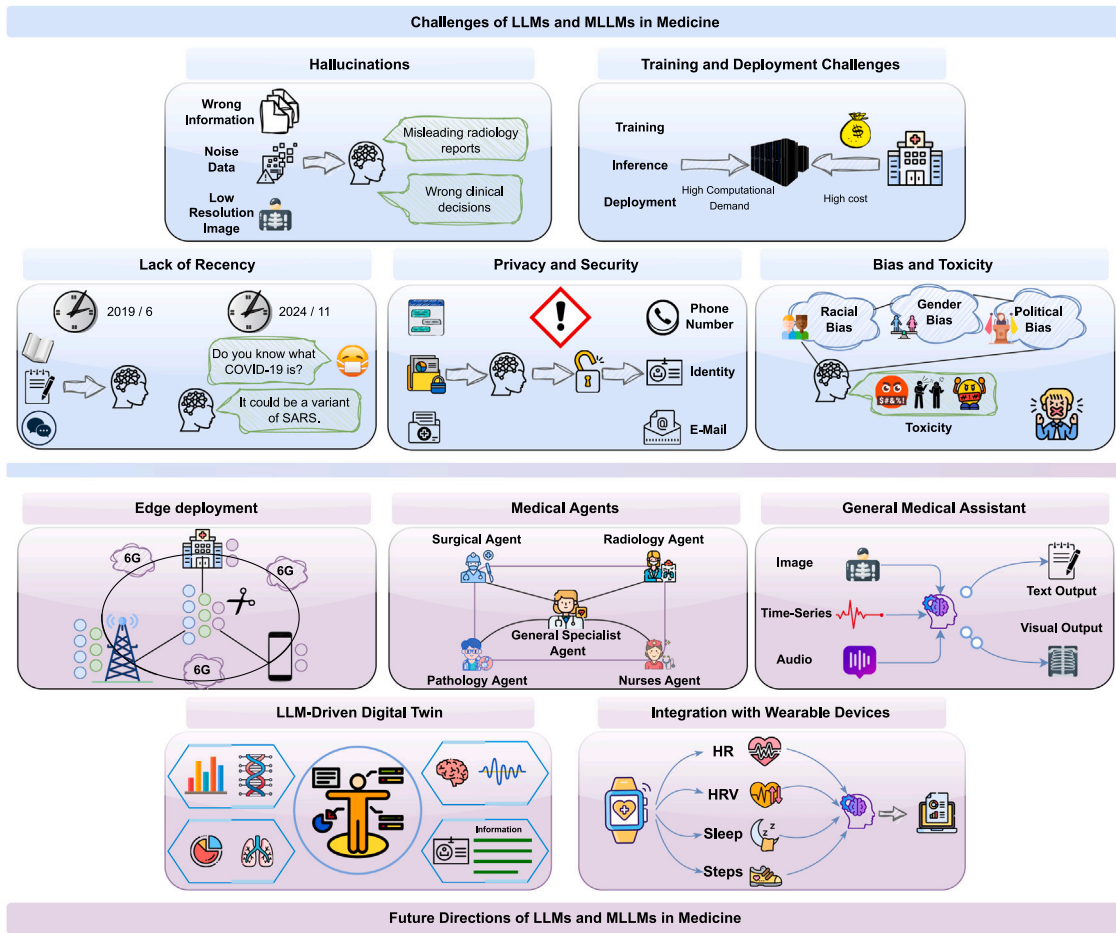


Fig. 11. Overview of challenges and future directions for medical LLMs and MLLMs in clinical settings.

7. Future directions of LLMs and MLLMs in medicine

This section examines future development trends in medical LLMs and MLLMs, envisioning their improved integration into clinical settings while offering forward-looking and insightful perspectives, as shown in Fig. 11.

7.1. Edge deployment

Due to their substantial computational and storage demands, existing LLMs and MLLMs are predominantly deployed on cloud servers. However, cloud-based deployment faces notable limitations in medical environments. First, in regions with poor network connectivity, communication latency with the cloud hinders the use of medical LLMs and MLLMs in critical clinical scenarios, such as surgical assistance. Second, transmitting data to cloud servers for cloud-based LLMs and MLLMs raises significant concerns about data privacy breaches. To address these challenges, deploying medical LLMs and MLLMs on edge devices is gaining traction as a promising future direction.

Edge deployment offers enhanced patient privacy protection and enables real-time responses [241]. Nevertheless, the limited computational capacity and memory of edge devices present significant challenges for deploying medical LLMs and MLLMs in such environments. Lin et al. [242] proposed a Mobile Edge Computing (MEC) architecture leveraging 6G networks to coordinate tasks between edge devices and servers as a solution to this challenge. For example, in scenarios demanding high privacy, complete device-side inference can be utilized to ensure full patient privacy protection. In more complex scenarios, this architecture supports dynamic adjustments of model splitting and task

allocation. For instance, in surgical assistance, the MEC architecture can deploy lightweight model components on edge devices to enable real-time responses, while offloading computationally intensive tasks to nearby edge servers. This setup supports distributed collaborative inference through 6G communication, reducing the burden on edge devices.

7.2. Medical agents

Clinical diagnosis is a multidisciplinary process requiring collaboration among experts from fields such as pathology, radiology, and surgery. LLM- and MLLM-based agents can simulate departmental experts, facilitating team collaboration to deliver comprehensive and accurate diagnostic support [15].

For example, when handling a complex medical case, a general practitioner agent can decompose the task into clear sub-tasks based on the patient's specific conditions and requirements, dynamically assigning them to the most suitable expert agents. Each expert agent provides analysis specific to their assigned sub-tasks. A radiology agent analyzes imaging studies to generate a radiology report, a surgical agent develops a surgical plan, and a pathology agent examines tissue samples or blood test results to provide cellular- or molecular-level diagnostic insights. After each agent completes their assigned sub-tasks, a collaborative mechanism consolidates their diagnostic inputs into a comprehensive medical opinion, which is then communicated back to the general practitioner agent [243,244]. Additionally, during agent collaboration, shared data and insights contribute to improved overall diagnostic recommendations. However, effective collaboration among multiple agents requires low-latency communication, especially

in urgent scenarios like surgical assistance, where real-time interaction is essential. Consequently, 6G communication technology can also offer vital technical support for enabling seamless multi-agent collaboration.

7.3. General medical assistant

Although medical MLLMs currently support visual and text modalities, enabling tasks such as medical image report generation and visual question answering, clinical environments demand these models to process additional input modalities, including time series and audio data [245] and audio data. These additional input modalities can provide more comprehensive patient information, particularly playing a crucial role in dynamic monitoring and real-time diagnosis.

At present, medical LLMs and MLLMs primarily rely on static data, such as medical records and imaging studies, limiting their capacity to account for patients' dynamic data. Incorporating time series data allows medical MLLMs to analyze health trends and detect early signs of disease progression, leading to more precise alerts and predictions. For example, in an ICU, in an ICU, medical MLLMs can integrate continuous ECG monitoring and blood pressure variations to predict cardiac arrest risks and recommend timely interventions. Similarly, audio data, such as cough recordings, can aid in diagnosing lung diseases or predicting respiratory failure risks. Voice analysis by medical MLLMs can also assess a patient's emotional state, supporting psychiatrists in diagnosing mental health conditions like depression and anxiety [246–248].

Beyond additional input modalities, medical MLLMs are also expected to support a broader range of output modalities. Currently, most medical MLLMs rely on LLM components as output modules, restricting outputs to text. In clinical practice, medical MLLMs require visual output modules for region- and pixel-level outputs, such as segmenting pathological areas in images according to physician instructions to deliver more detailed diagnostic evidence.

7.4. LLM-driven digital twin

Clinical trials are essential research methods for evaluating the safety and efficacy of new drugs, medical devices, treatment methods, or interventions in human subjects. While these trials are vital for medical advancements, they often require thousands of participants and take several years to complete, resulting in significant costs. In recent years, AI models based on EHRs have been proposed as an economical alternative to traditional clinical trials, aiming to predict clinical trial outcomes [249,250]. However, these models frequently overlook individual patient differences and disease variability, resulting in discrepancies between predicted and actual outcomes [251].

To address this challenge, researchers have suggested leveraging digital twin technology to simulate patients' physiological states, disease progression, and treatment effects [251–254]. This technology combines historical data with real-time information, offering doctors more accurate and comprehensive personalized health insights [252]. However, current digital twin technologies often rely on limited patient data and lack adequate real-world contextual knowledge, posing challenges for accurate clinical trial outcome prediction [250].

Combining LLMs with digital twin technology offers the potential to create more advanced and patient-centered intelligent twins. Through training on large-scale datasets, LLMs acquire extensive real-world knowledge and common sense. Integrating LLMs with digital twins can inject a stronger real-world context, reasoning ability, and interactive capability into the digital twin system, making it more aligned with individual patient characteristics. Consequently, LLM-driven digital twins facilitate accurate predictions of patients' physiological and pathological statuses during clinical trials while considering individual variations and disease complexities [252–254]. Additionally, by utilizing the generative capabilities of LLMs, this technology can produce high-fidelity

trial data tailored to specific patients, effectively addressing the issue of limited medical data while safeguarding patient privacy.

In conclusion, LLM-driven digital twin technology could significantly reduce the time and costs associated with clinical trials while maintaining patient privacy and security. It is expected to become an essential support and technological foundation for future medical research.

7.5. Integration with wearable devices

Wearable sensors have become an essential component of modern technology, enabling real-time monitoring of individual behavior and health through integration with devices such as smartwatches [255, 256]. This technology provides novel opportunities for the early detection of health issues. However, most current wearable devices are limited to monitoring physiological indicators and lack the capacity for in-depth data analysis and interpretation, constraining their ability to deliver truly personalized health advice.

In the future, medical MLLMs expected to evolve toward being smaller in scale, higher in performance, and incorporating more modalities, presenting significant opportunities for seamless integration with wearable devices. The integration of MLLMs into wearable devices could greatly enhance their intelligence and improve user interaction. For example, MLLMs could facilitate natural language interactions, enabling users to inquire about their physiological state in real-time and receive accurate, easily understandable responses [257]. Furthermore, MLLMs can combine physiological data collected by the device to provide highly personalized health recommendations. For instance, if a user's physiological indicators suggest fatigue, the device could intelligently prompt them to adjust their schedule, take more rest, and extend sleep duration. Additionally, MLLMs could analyze sleep data, generate detailed sleep quality reports, and offer targeted recommendations for improvement [258,259]. This functionality not only enhances users' understanding of their health but also promotes personalized and evidence-based health management in daily life.

In conclusion, integrating MLLMs with wearable devices could not only enhance their functionality but also lay the groundwork for personalized health assistants. This integration model is anticipated to offer users a more intelligent and user-friendly health management experience, propelling digital healthcare technologies to new heights.

8. Conclusion

In recent years, advancements in LLMs have driven significant breakthroughs in NLP, enabling researchers to make substantial progress toward artificial general intelligence by extending LLMs into the multimodal domain, resulting in the creation of MLLMs. Concurrently, the rapid development and impressive performance of general LLMs and MLLMs have spurred the emergence of numerous medical LLMs and MLLMs. This survey aims to help researchers and medical practitioners understand the technological advancements and developmental status of medical LLMs and MLLMs. It focuses on the paradigm shift of LLMs and MLLMs, highlighting their evolution from feature engineering to structure engineering, objective engineering, and now to prompt engineering and data engineering. The survey summarizes the mainstream architectures of current LLMs and MLLMs, compiles a list of existing medical LLMs and MLLMs, and provides insights into various architectures and model components. Additionally, it presents a comprehensive guide on existing medical datasets, model fine-tuning approaches, and evaluation methods, aiding researchers and practitioners in building medical LLMs and MLLMs. The survey also examines the broad potential of LLMs and MLLMs in diverse clinical applications, such as medical diagnosis, clinical report generation, medical education, mental health services, and surgical assistance. Despite their remarkable achievements, medical LLMs and MLLMs face several critical challenges and limitations that hinder their practical

deployment in clinical settings. Consequently, this survey addresses these challenges, including hallucinations, training and deployment issues, lack of recency, privacy and security, bias and toxicity, while offering potential solutions to support the practical application of future medical LLMs and MLLMs. Lastly, the survey explores future directions for medical LLMs and MLLMs, such as edge deployment, medical agents, general medical assistant, LLM-driven digital twin, and integration with wearable devices, offering forward-looking and insightful analysis. In conclusion, this survey delivers a comprehensive analysis of medical LLMs and MLLMs, covering their background, principles, applications, challenges, and future directions, with the aim of advancing their development in clinical medicine and fostering AI integration in healthcare.

CRediT authorship contribution statement

Hanguang Xiao: Writing – review & editing. **Feizhong Zhou:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Xingyue Liu:** Investigation. **Tianqi Liu:** Investigation. **Zhipeng Li:** Investigation. **Xin Liu:** Investigation. **Xiaoxuan Huang:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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